The Bias-Variance Tradeoff Determines Benefit of Ensemble Learners versus Single-Model Learners on Clustered Data

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Abstract

We consider covariate data that naturally partition into clusters and compare two learning strategies: (i) ensembling separate learners trained on each cluster, and (ii) training a single global model on all clusters, under the assumption that the covariate–outcome relationship is homogeneous across clusters. We analyze how the bias–variance trade-off of the underlying learning algorithms mediates the potential gains from ensembling versus merging. Specifically, we contrast two algorithms with opposite bias–variance regimes under our chosen data-generating mechanisms: Least Squares regression under a linear model, where variance dominates the mean squared error (MSE); and Random Forests under a measurable and square integrable target function, where we demonstrate that squared bias dominates the MSE. Our results show that ensembling unbiased least squares regressors trained on clusters is asymptotically less effective than fitting a single regression on all data. By contrast, ensembles of forests substantially improve over a single forest in a clustered setting. We attribute the advantage of clustering-based ensembling to bias reduction, indicating that such ensembling can be advantageous for base learners in which bias dominates MSE even when the outcome model remains homogeneous across clusters.

Keywords: Machine Learning, Ensemble Learning, High Dimensional Regression, Linear Regression, Random Forest, Statistical Theory

1. Introduction

Natural clusters and batch effects are a common data artifact across a variety of applications, motivating the need for prediction algorithms that are robust to heterogeneity in the distribution of the features and capable of generalizing (Goh et al., 2017; Chauhan et al., 2010). Ensembling frameworks have been previously explored in such a setting and shown to generalize more effectively than single-model learners (Patil and Parmigiani, 2018; Deodhar and Ghosh, 2007; Trivedi et al., 2015; Ramchandran et al., 2020). In this paradigm, datasets are first separated into their component clusters or natural partitions, a learning algorithm is then trained on each cluster, and finally the predictions made by each single-cluster learner are combined using weighting approaches that reward cross-cluster prediction ability within the training set - as a shorthand, we will henceforth refer to this approach simply as 'ensembling'. This is in contrast to what we term 'merging', which represents the predictions made by a single model trained using the chosen learning algorithm on the entire training dataset. Learning algorithms that have been explored within this framework include Neural Networks, Random Forest, and regularized regression (Patil and Parmigiani, 2018; Guan et al., 2019; Ramchandran et al., 2020).

In this paper, we investigate why this ensembling framework is beneficial for some learning algorithms when applied to clustered data and not beneficial for others. To isolate the effect of the cluster structure of the features on the accuracy of ensembling versus merging, we consider only the scenario in which there is one global function defining the relationship between the features

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and the outcome; that is, the functional form of the outcome does not vary across clusters. We posit that the bias-variance tradeoff of the underlying learning algorithm mediates the benefit of ensembling: learning algorithms for which the bias dominates the MSE are effective within the ensembling framework, while learning algorithms for which the variance dominates the MSE are more effective within the merging framework. To support this claim, we analyze Random Forests and Least Squares regressors as the base algorithms representing each extreme of the bias-variance tradeoff given our assumptions on the data-generating process, and present theoretical limits for the asymptotic risk of ensembling versus merging in each case. We chose these two learning approaches for our analysis due to empirical support in the previous work referenced that ensembling is effective for Random Forest learners but not for Least Squares regression on clustered data, the ubiquity of both methods in practical applications, and the analytical tractability of both approaches given our assumptions on the data.

This paper is organized as follows: we first present our analysis using Random Forest learners on clustered data where the clusters are drawn from non-overlapping uniform distributions, building upon previous analytical work by Biau (2012) and Klusowski (2020) on characterizing the bias and variance of Random Forest learners for uniform data. Of note, we relax the assumptions in earlier work that the outcome model is linear, allowing for more complex models including piecewise-constant models that exactly match the training objective of decision trees. In this way, we analyze under a class of functions including those for which Random Forest is well specified. We calculate the limiting ratio of the ensembling risk versus merging risk for when the data contains two or more clusters, and find that ensembling is strictly more accurate than merging in this case. We next analyzing the prediction performance for ensembles of Least Squares linear regression learners when there are two clusters in data drawn from gaussian distributions, assuming a linear outcome to ensure unbiased estimation. We show that regardless of whether the dimension is fixed or increases proportionally to the sample size, merging is in fact asymptotically strictly better than ensembling.

The distributional assumptions in each analysis are chosen primarily for analytical tractability. Nonetheless, comparing these complementary formulations allows us to draw connections between them and characterize broader principles governing bias–variance trade-offs across learning paradigms. In the discussion, we consider how these results may persist under more general conditions. The discussion elaborates on potential relaxations of these assumptions and implications for other learning approaches exhibiting similar bias–variance trade-offs. We also consider the broader implications for when cluster-aware ensembling is likely to be most effective in practice.

2. Notation

We denote the training data set as $\mathcal{D}_n = (\mathbb{X}, \mathbb{Y}) \in (\mathbb{R}_{n\times S}, \mathbb{R})$, where S represents the number of covariates in the training set and n the training sample size. The training dataset can be further expressed as $(\mathbb{X}, \mathbb{Y}) = \left([\mathbb{X}_1 \dots \mathbb{X}_k]^T, [\mathbb{Y}_1 \dots \mathbb{Y}_k]\right)$, where k is the number of clusters and $(\mathbb{X}_b, \mathbb{Y}_b) \in (\mathbb{R}_{n_b\times S}, \mathbb{R})$ is training data corresponding to the bth cluster for $b = 1, \dots, k$. We specify that the clusters are equally sized with $n_b = n/k$ samples per cluster, and that all samples are independent of one another.

We shall utilize the following language convention and model definitions for the theoretical analysis. Single Cluster Learners (SCL's) refer to an algorithm that produces a prediction model using a single cluster; in this paper, we consider Least Squares linear regression and Random Forests as SCL's. The term *Ensemble* will indicate training an SCL on each cluster within the training set

and then combining all cluster-level predictions using a specific weighting strategy, creating a single predictor that can be applied to external studies, as follows:

Algorithm 1: General framework for training the *Ensemble*

- Train an SCL on $(\mathbb{X}_b, \mathbb{Y}_b)$ for each cluster $b = 1, \ldots, k$. The predictions of each SCL on a test point $\mathbf{x}_\star = \hat{Y}_b(\mathbf{x}_\star; \theta, \mathcal{D}_n)$
- Obtain predictions for the *Ensemble* on test point $\mathbf{x}_\star$ by weighting the predictions of each SCL; $\hat{Y}_E(\mathbf{x}_\star; \theta, \mathcal{D}_n) = \sum_{b=1}^{k} w_b \hat{Y}_b(\mathbf{x}_\star; \theta, \mathcal{D}_n)$ for $b = 1, \ldots, k$ and SCL-specific weights $w_1, \ldots, w_k$

The term *Merged* will refer to the strategy of training the same single learning algorithm chosen to create the SCL's on the entire training dataset. The prediction made by the *Merged* learner is denoted as $\hat{Y}_M(\mathbf{x}_\star; \theta, \mathcal{D}_n)$. All specifications for this general setup will be discussed in the relevant sections.

3. Analyzing Random Forest Learners

In this section, we examine the asymptotic risk of ensembles of Random Forest learners compared to a single Random Forest on clustered data, showing that bias reduction is the primary mechanism through which the general cluster-based ensembling framework produces improvements for forest-based learners.

3.1. Analytical setup

In this analysis, we specify that the training covariate set consists of k equally sized, non-overlapping, uniform clusters spanning the $[0,1]^S$ hypercube: $\mathbb{X}_1 \overset{\text{i.i.d.}}{\sim} \left[\mathrm{U}\left(0, \frac{1}{k}\right)\right]^S, \mathbb{X}_2 \overset{\text{i.i.d.}}{\sim} \left[\mathrm{U}\left(\frac{1}{k}, \frac{2}{k}\right)\right]^S, \ldots, \mathbb{X}_k \overset{\text{i.i.d.}}{\sim} \left[\mathrm{U}\left(\frac{k-1}{k}, 1\right)\right]^S$. We can therefore represent each training sample as $(\mathbf{x}_i, Y_i) \overset{\text{i.i.d.}}{\sim} \left(\sum_{b=1}^{k} A_i \left[\mathrm{U}\left(\frac{b-1}{k}, \frac{b}{k}\right)\right]^S, f(\mathbf{x}_i)\right)$, where $A_i \sim Multinomial(k, n/k)$ is a multinomial random variable representing the cluster assignment of $\mathbf{x}_i$. The outcome $Y_i = f(\mathbf{x}_i)$, where we assume that $f(\cdot)$ is measurable and square-integrable. Test point $(\mathbf{x}_\star, Y_\star)$ is drawn from the same distribution as the training set.

3.1.1. CENTERED RANDOM FORESTS

We define a Random Forest as an ensemble of base regression trees $\{f_n(\mathbf{x}_\star; \theta_m, \mathcal{D}n), m \geq 1\}$, where θ_m are i.i.d. copies of a randomizing variable θ determining node splits. The forest-level prediction is $\bar{f}_n(\mathbf{x}_\star; \theta, \mathcal{D}_n) = \mathbb{E}_\theta\left[f_n(\mathbf{x}_\star; \theta, \mathcal{D}_n)\right]$, typically estimated by averaging predictions from multiple trees trained on bootstrap samples (Breiman, 2001). In this section, we analyze the Centered Random Forest method proposed in Breiman (2004), which simplifies the original algorithm by assuming θ is independent of the training sample D_n. This excludes bootstrapping and data-dependent tree-building strategies but permits θ to be based on a second sample $\mathcal{D}_n' \overset{\text{i.i.d.}}{\sim} \mathcal{D}_n$, enabling the splits to be optimized in a manner closely resembling how they would have been with the actual training set. We build upon the analytical work to characterize this model in Klusowski (2020) and Biau (2012), keeping our notation consistent whenever possible.

Centered Random Forest trees are constructed over $\log_2 c_n$ iterations, where $c_n \geq 2$ is a user-defined parameter that possibly depends on n. Each tree is built as follows: at each node, a coordinate of $\mathbf{x}$ is selected, with the jth feature having a probability $p_{nj} \in (0, 1)$ of being selected, and splits occur at the midpoint of the chosen coordinate. Following from Klusowski (2020), $p_{nj} \to 1/S$ as $n \to \infty$. Each tree outputs the average Y_i for $\mathbf{x}_i$ in the same leaf node partition as $\mathbf{x}_\star$, denoted $A_n(\mathbf{x}_\star, \theta)$. The midpoint splitting is for mathematical convenience; more complex mechanisms could be used with $\mathcal{D}_n^{'}$, but would complicate the analysis without providing additional insight into ensembling strategies. The individual tree predictor can therefore be represented as $f_n(\mathbf{x}_\star; \theta, \mathcal{D}_n) = \frac{\sum_{i=1}^n Y_i \mathbb{1}\{\mathbf{x_i} \in A_n(\mathbf{x}_\star, \theta)\}}{\sum_{i=1}^n \mathbb{1}\{\mathbf{x_i} \in A_n(\mathbf{x}_\star, \theta)\}} \mathbb{1}\{\epsilon_n(\mathbf{x}_\star, \theta)\}$, where $\epsilon_n(\mathbf{x}_\star, \theta)$ is the event that $\sum_{i=1} \mathbb{1}\{\mathbf{x_i} \in A_n(\mathbf{x}_\star, \theta)\} > 0$; that is, that there is at least one training point that falls within the same leaf node as $\mathbf{x}_\star$. We can then obtain the expected prediction made by the Centered Random Forest by taking the expectation of the individual tree predictors with respect to the randomizing variable θ: $\bar{f}_n(\mathbf{x}_\star; \theta, \mathcal{D}_n) = \sum_{i=1}^n Y_i \mathbb{E}_\theta \left[\frac{\mathbb{1}\{\mathbf{x_i} \in A_n(\mathbf{x}_\star, \theta)\}}{\sum_{i=1}^n \mathbb{1}\{\mathbf{x_i} \in A_n(\mathbf{x}_\star, \theta)\}} \mathbb{1}\{\epsilon_n(\mathbf{x}_\star, \theta)\} \right]$

For the *Ensemble*, we use Centered Random Forests as the SCL in the procedure described in Algorithm 1, and cluster-specific weights $w_1, \ldots, w_k = 1$. Creating the ensemble by adding the predictions of each forest is optimal due to the data setup, since we specify non-overlapping clusters. Note that Centered Random Forests do not extrapolate beyond their training data, so $\hat{Y}_b(\mathbf{x}_\star) = 0$ if the cluster assignment of $\mathbf{x}_\star \neq b$. The strategy of adding each component forest in the Ensemble is analogous to having each cluster-specific forest only predict on test points arising from the same cluster as it was trained on, given our assumption that the clusters are non-overlapping. This setup is chosen for tractability while capturing the comparison of interest; we outline how these results can be extended to non-overlapping and non-uniform clusters in the discussion section.

The aim of the following section is to derive the limit of the ratio $\frac{MSE(\hat{Y}_E(\mathbf{x}_\star))}{MSE(\hat{Y}_M(\mathbf{x}_\star))} = \frac{\mathbb{B}_E^2 + \mathbb{V}_E}{\mathbb{B}_M^2 + \mathbb{V}_M}$, where we define $\mathbb{B}_E$ and $\mathbb{V}_E$ as the bias and variance of the *Ensemble* predictions and $\mathbb{B}_M$ and $\mathbb{V}_M$ as the analogous quantities for the *Merged*.

3.2. Asymptotic Analysis

In Lemma 1, we show that the *Ensemble* and *Merged* predictions can be represented as weighted averages of the training outcome values, with the choice of weights as the differentiating factor between the methods.

Lemma 1

(i) For data divided into k clusters, the predictions of the k cluster-level forests within the Ensemble learner on new point $\mathbf{x}_\star$ can be expressed as

$$\hat{Y}_b(\mathbf{x}_\star; \theta, \mathcal{D}_n) = \sum_{i=1}^n Y_i \mathbb{E}_\theta \left[W_{ib}(\mathbf{x}_\star, \theta) \right]$$

where

$$W_{ib}(\mathbf{x}_\star, \theta) = \frac{\mathbb{1}_{\{\mathbf{x_i} \in A_n(\mathbf{x}_\star, \theta)\}} \mathbb{1}_{\{\mathbf{x}_i \in [\frac{b-1}{k}, \frac{b}{k}]^S\}} \mathbb{1}_{\{\mathbf{x}_\star \in [\frac{b-1}{k}, \frac{b}{k}]^S\}}}{N_b(\mathbf{x}_\star, \theta)} \times \mathbb{1}_{\{\epsilon_{nb}(\mathbf{x}_\star, \theta)\}}$$

$$N_b(\mathbf{x}_\star, \theta) = \sum_{i=1}^n \mathbb{1}_{\{\mathbf{x_i} \in A_n(\mathbf{x}_\star, \theta)\}} \mathbb{1}_{\mathbf{x}_i \in [\frac{b-1}{k}, \frac{b}{k}]^S\}} \mathbb{1}_{\mathbf{x}_\star \in [\frac{b-1}{k}, \frac{b}{k}]^S\}}$$

and $\epsilon_{nb}(\mathbf{x}_\star, \theta)$ *is the event that* $N_b(\mathbf{x}_\star, \theta) > 0$ *for* $b = 1, \ldots, k$*;* $N_b(\mathbf{x}_\star, \theta)$ *represents the number of total training points from* $\mathbb{X}_b$ *that fall into the same partition of the test point* $\mathbf{x}_\star$ *given that* $\mathbf{x}_\star \in [\frac{b-1}{k}, \frac{b}{k}]^S$

Specifically, the predictions of the overall Ensemble as

$$\begin{aligned}\hat{Y}_E(\mathbf{x}_\star; \theta, \mathcal{D}_n) &= \sum_{i=1}^{n} \sum_{b=1}^{k} Y_i \mathbb{E}_\theta \left[W_{ib}(\mathbf{x}_\star, \theta) \right] \\ &= \sum_{i=1}^{n} Y_i \mathbb{E}_\theta \left[W_i(\mathbf{x}_\star, \theta) \right]\end{aligned}$$

where $W_i(\mathbf{x}_\star, \theta) = \sum_{b=1}^{k} W_{ib}(\mathbf{x}_\star, \theta)$.

(ii) The predictions of the Merged learner on new point $\mathbf{x}_\star$ *can be expressed as*

$$\hat{Y}_M(\mathbf{x}_\star; \theta, \mathcal{D}_n) = \sum_{i=1}^{n} Y_i \mathbb{E}_\theta \left[H_i(\mathbf{x}_\star, \theta) \right]$$

where $H_i(X, \theta) = \frac{\mathbb{1}_{\{\mathbf{x_i} \in A_n(\mathbf{x}_\star, \theta)\}}}{N_n(\mathbf{x}_\star, \theta)} \mathbb{1}_{\{\epsilon_n(\mathbf{x}_\star, \theta)\}}$ *and* $N_n(\mathbf{x}_\star, \theta) = \sum_{i=1}^{n} \mathbb{1}_{\{\mathbf{x_i} \in A_n(\mathbf{x}_\star, \theta)\}}$*, the number of total training samples falling into the same box as* $\mathbf{x}_\star$*. Finally,* $\epsilon_n(\mathbf{x}_\star, \theta)$ *is the event that* $N_n(\mathbf{x}_\star, \theta) > 0$.

To develop theoretical results regarding the performance of these two approaches, we build on Klusowski (2020) in characterizing the asymptotic behavior of the bias and variance components of the MSE for Centered Random Forests. We extend upon these results to allow for clusters in the training and test data and to remove any requirement on the functional form of the outcome model $f(\mathbf{x})$. We note that the latter represents a significant step forward upon previous analytical work on Random Forests, which typically assume a linear or piecewise linear outcome model.

We begin by stating supplementary results that support the primary performance comparison between the *Ensemble* and the *Merged* in Theorem 2 below. First, we can show that the variance terms $\mathbb{V}_E$ and $\mathbb{V}_M \to 0$ at a rate of $O\left(\frac{1}{n}\right)$, the formal statement and proof of which we provide in Lemma 7 in the Appendix. This follows directly from the results in [Klusowski (2020), Proposition 2] and its corresponding proof. We further can show that the squared bias terms ${\mathbb{B}_\mathbb{E}}^2$ and ${\mathbb{B}_\mathbb{M}}^2$ converge to 0 at a rate greater $O\left(\frac{1}{n}\right)$ given minimal assumptions on S and c_n, the proof of which is given in Remark 6 in the Appendix. Therefore, we can characterize the limit of $\frac{MSE(\hat{Y}_E(\mathbf{x}_\star))}{MSE(\hat{Y}_M(\mathbf{x}_\star))}$ by the limit of $\frac{\mathbb{B}_E^2}{{\mathbb{B}_\mathbb{M}}^2}$; that is, the bias terms dominate the RMSE as $n \to \infty$. To further characterize this ratio, we build upon [Klusowski (2020), Theorem 4], from which we can express the leading term of $\mathbb{B}_E$ as $\sqrt{n(n-1)\mathbb{E}_{\mathbf{x}_\star, \mathcal{D}_n, \beta}\left[\mathbb{E}_\theta\left[W_1\right]\left(\hat{Y}_1(\mathbf{x}_\star) - f(\mathbf{x}_\star)\right)\mathbb{E}_\theta\left[W_2\right]\left(\hat{Y}_2(\mathbf{x}_\star) - f(\mathbf{x}_\star)\right)\right]}$ and the leading term of $\mathbb{B}_M$ as $\sqrt{n(n-1)\mathbb{E}_{\mathbf{x}_\star, \mathcal{D}_n, \beta}\left[\mathbb{E}_\theta\left[H_1\right]\left(\hat{Y}_1(\mathbf{x}_\star) - f(\mathbf{x}_\star)\right)\mathbb{E}_\theta\left[H_2\right]\left(\hat{Y}_2(\mathbf{x}_\star) - f(\mathbf{x}_\star)\right)\right]}$ for weights W_1, W_2, H_1, and H_2 as defined in Lemma 1. We can then derive the limit of the performance ratio between the *Ensemble* and the *Merged* as presented in Theorem 2 below.

Theorem 2 *Given a training set with n samples, S covariates, and $k \geq 2$ equally-sized uniform clusters $\{(\mathbf{x}_i, Y_i) \overset{\text{i.i.d.}}{\sim} \left(\sum_{b=1}^{k} A_{ib} \left[\mathrm{U}\left(\frac{b-1}{k}, \frac{b}{k}\right)\right]^S, f(\mathbf{x}_i)\right) : i = 1, \ldots, n\}$, where $A_i \sim Multinomial(1, [k/n, \ldots, k/n] \in \mathbb{R}_k)$ is a multinomial random variable representing the cluster assignment of $\mathbf{x}_i$ and $f(x)$ is a measurable and square integrable function, we can represent the predictions of the Ensemble and the Merged on test point $\mathbf{x}_\star \overset{\text{i.i.d.}}{\sim} \sum_{b=1}^{k} A_\star \left[\mathrm{U}\left(\frac{b-1}{k}, \frac{b}{k}\right)\right]^S$ as $\hat{Y}_E$ and $\hat{Y}_M$, respectively. As $n \to \infty$, the ratio of the mean square errors corresponding to these predictions converges as follows:*

$$\lim_{n\to\infty} \frac{MSE(\hat{Y}_E(\mathbf{x}_\star))}{MSE(\hat{Y}_M(\mathbf{x}_\star))} \to \frac{1}{2}$$

Theorem 2 demonstrates that the asymptotic performance improvement of the *Ensemble* over the *Merged* is independent of the number of true clusters k for $k \geq 2$. To understand this phenomenon, consider that ensembling involves a trade-off: it reduces the number of samples used to train each learner (decreasing efficiency compared to the *Merged*), but it also allows each learner to focus on regions of the covariate space that better reflect the distribution of the test point, thereby improving accuracy. The proof of Theorem 2 decomposes the ratio $\frac{MSE(\hat{Y}_E(\mathbf{x}\star))}{MSE(\hat{Y}_M(\mathbf{x}\star))}$ into two components that capture this trade-off. It shows that when these components are multiplied, terms involving k cancel out, resulting in a final ratio of $\frac{1}{2}$. Although each learner in the *Ensemble* is trained on $\frac{1}{k}$ times the number of samples as the *Merged*, the benefit of the *Ensemble* approach lies in its ability to more effectively parse out the heterogeneity introduced by clusters from the true covariate-outcome relationship. This advantage outweighs the loss in efficiency due to the reduced sample size as $n \to \infty$. To our knowledge, this work provides the first theoretical characterizations of the Random Forest algorithm that account for heterogeneous data without imposing a linear functional form on the outcome model $f(\mathbf{x})$.

Figure 1 presents simulation results illustrating Theorem 2's relevance to clustered data generated from three distributional paradigms: uniform, Gaussian, and Laplace. We use the traditional Random Forest algorithm for training instead of the Centered Forest algorithm to assess the ability of the latter to approximate the behavior of the former. The uniform distribution simulation matches our theoretical setup, with non-overlapping clusters of width $\frac{1}{k}$. In contrast, the Gaussian and Laplace simulations feature overlapping clusters characterized by different location parameters and identity scale parameters. Across all distributions and the number of simulated clusters, the ratio of *Ensemble* to *Merged* MSEs is consistently close to $\frac{1}{2}$; as expected, the uniform simulation exhibits a high consistency with the theory. When assumptions are violated, in the Gaussian and Laplace cases, ratios are close to $\frac{1}{2}$ as k gets large, but can deviate for smaller values. This illustrates that our analytical framework effectively captures the performance improvements of the *Ensemble* across diverse and realistic scenarios at least for large k.

3.3. Analyzing Linear Least Squares Regression

In this section, we examine the asymptotic risk of ensembles of Least Square linear regressions compared to a single linear regression on clustered data under a linear outcome model, for both fixed and high dimensional data. We show that the MSE of the *Merged* (which is entirely characterized by the variance) is strictly better than that of the *Ensemble*. When the bias does not dominate the

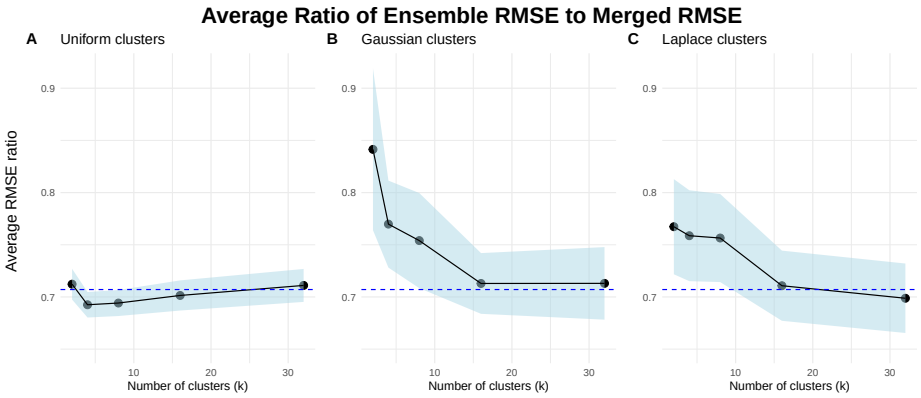


Figure 1: Average ratio of *Ensemble* RMSE to *Merged* RMSE as a function of the number of clusters in the training set (n = 5,000). The outcome is simulated as a linear combination of 20 covariates, out of which 10 have zeroed out coefficients to induce sparsity, with added Gaussian noise. The means (points) and 95% CIs (shaded blue regions) are calculated over 150 replicates at each k. The dotted line is at the theoretical limiting ratio $1/\sqrt{2}$. **(A)** Uniform clusters **(B)** Multivariate Gaussian clusters **(C)** Multivariate Laplace clusters

MSE, we see that there is no benefit for ensembling when there is no heterogeneity in the outcome model across clusters.

3.3.1. ANALYTICAL SETUP

We assume that the distribution of each training cluster is given by $\mathbb{X}_b \overset{\text{i.i.d.}}{\sim} \text{MVN}_S(\boldsymbol{\mu}_\mathbf{b}, \text{I})$ for $b = 1, \ldots, K$ and the distribution of the test point $\mathbf{x}_\star \overset{\text{i.i.d.}}{\sim} \text{MVN}_S(\mathbf{0}, \text{I})$. Note that unlike in the Random Forest analysis, here we assume that the test point follows a standard normal distribution, whereas the training samples can be drawn from distributions with non-zero means. This choice is made for a couple key reasons; first, it allows the expressions derived within this section to have closed-form solutions, and second, it allows us to model a greater degree of covariate shift, where we have both internal heterogeneity in the distribution of the training covariates and shift between the distributions of training and test samples. We assume a true linear model for the outcome, $f(x) = x^T\beta$, for $\beta, x \in \mathbb{R}_{S\times 1}$, and denote the vector of true outcomes for cluster b and the entire training dataset as $\mathbb{Y}_b$ and $\mathbb{Y}$, respectively. Note that unlike in the Random Forest analysis, we do not have to place any stipulations on the sample size of each cluster as long as the total sample size of the training data equals n.

The *Merged* and *Ensemble* strategies for linear least squares regression SCL's both produce unbiased predictions, regardless of weighting strategy. We begin by discussing the generalizable weighting approach used to create the *Ensemble*: Inverse Variance Weighting (IVW). Given test point $\mathbf{x}_\star$, we can actually compute weights that are proportional to the inverse variance of each learner when predicting on this new point, making use of the functional form of each SCL's variance, $\text{Var}(\hat{Y}_b(\mathbf{x}_\star))$ for $b = 1, \ldots, k$, in the calculations. This requires that the ensemble be built only after the test point's features are known. In this case, the weights yielded by inverse variance weighting

are equivalent to

$$\gamma_b = \left[\sum_{b=1}^{k}\left[\mathbf{x}_\star^T(\mathbb{X}_b^T\mathbb{X}_b)^{-1}\mathbf{x}_\star\right]^{-1}\right]^{-1}\left[\mathbf{x}_\star^T(\mathbb{X}_b^T\mathbb{X}_b)^{-1}\mathbf{x}_\star\right]^{-1}$$

for $b = 1, ..., k$. Therefore, in order to predict on new point $\mathbf{x}_\star$, the weights given to each SCL within the ensemble must be inversely proportional to the variance of each SCL-level prediction on $\mathbf{x}_\star$. The *Ensemble* and *Merged* predictions, denoted by $\hat{Y}_E(\mathbf{x}_\star)$ and $\hat{Y}_M(\mathbf{x}_\star)$ respectively, are then given by

$$\hat{Y}_E(\mathbf{x}_\star) = \left[\sum_{b=1}^{k}\left[\mathbf{x}_\star^T(\mathbb{X}_b^T\mathbb{X}_b)^{-1}\mathbf{x}_\star\right]^{-1}\right]^{-1}\sum_{b=1}^{k}\underbrace{\left[\mathbf{x}_\star^T(\mathbb{X}_b^T\mathbb{X}_b)^{-1}\mathbf{x}_\star\right]^{-1}}_{\text{Var}(\hat{Y}_b)}\times\underbrace{\mathbf{x}_\star^T(\mathbb{X}_b^T\mathbb{X}_b)^{-1}\mathbb{X}_b^T\mathbb{Y}_b}_{\hat{Y}_b}$$

$$\hat{Y}_M(\mathbf{x}_\star) = \mathbf{x}_\star(\mathbb{X}^T\mathbb{X})^{-1}\mathbb{X}^T\mathbb{Y}$$

It is simple to see that both learners are unbiased, and therefore the MSEs are equal to the following variances:

$$\text{Var}\left[\hat{Y}_E(\mathbf{x}_\star)\right] = \left[\sum_{b=1}^{k}\left[\mathbf{x}_\star^T(\mathbb{X}_b^T\mathbb{X}_b)^{-1}\mathbf{x}_\star\right]^{-1}\right]^{-1} \tag{1a}$$

$$\text{Var}\left[\hat{Y}_M(\mathbf{x}_\star)\right] = \mathbf{x}_\star^T\left[\sum_{t=1}^{K}\mathbb{X}_b^T\mathbb{X}_b\right]^{-1}\mathbf{x}_\star \tag{1b}$$

3.3.2. Fixed dimensional asymptotics

Given the expressions in Equations 1a and 1b, we now incorporate the randomness of the training and test data in taking the limits, first assuming throughout this section that the dimension S is fixed as $n \to \infty$. We additionally specify that the number of clusters $k = 2$, and $\|\boldsymbol{\mu_1}\| = \|\boldsymbol{\mu_2}\| = 1$, in order to produce tractable limiting expressions. The following theorem formalizes the comparison between the asymptotic risks of the *Merged* and *Ensemble* given the above conditions.

Theorem 3 *We consider a training set with n samples, S covariates, and $k = 2$ Gaussian clusters $\{(\mathbf{x}_i, Y_i) \overset{\text{i.i.d.}}{\sim} \left(\sum_{b=1}^{2}\mathbb{1}\{\mathbf{x}_i \in \textit{Cluster } b\} \times \text{MVN}_S(\boldsymbol{\mu_b}, \text{I}), \mathbf{x}_i^T\beta\right) : i = 1, \ldots, n\}$, with $\beta \in \mathbb{R}_{S\times 1}$. We denote $\boldsymbol{\mu_b}$ as the cluster-specific means for $b = 1, 2$, and specify that $\|\boldsymbol{\mu_1}\| = \|\boldsymbol{\mu_2}\| = 1$. We denote $\lambda_b = \frac{n_b}{n}$ for $b = 1, 2$; by construction, $\lambda_1 + \lambda_2 = 1$. Then, we can represent the predictions of the Ensemble and the Merged on test point $\mathbf{x}_\star \overset{\text{i.i.d.}}{\sim} \text{MVN}_S(\mathbf{0}, \text{I})$ as $\hat{Y}_E$ and $\hat{Y}_M$, respectively. As $n \to \infty$, the ratio of the mean square errors corresponding to these predictions converges as follows:*

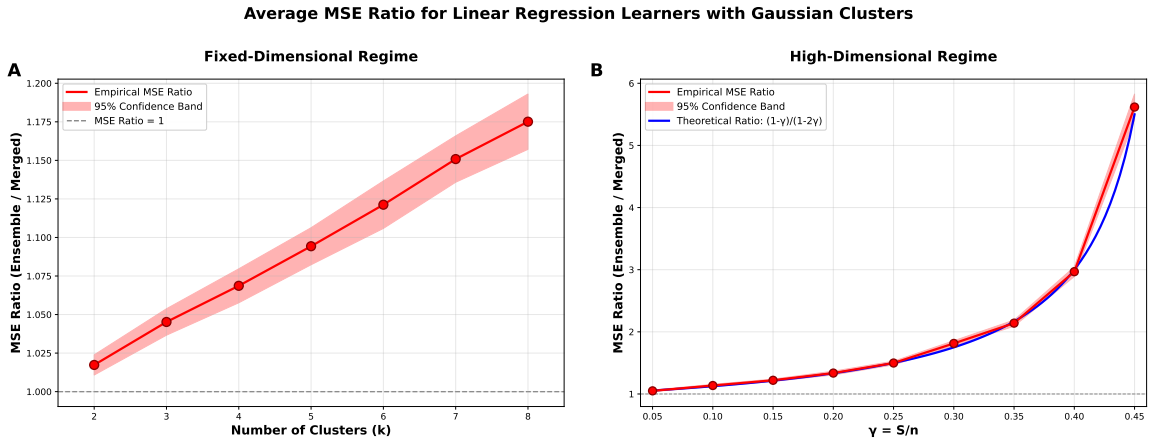


Figure 2: Ratio of *Ensemble* MSE to *Merged* MSE for linear regression learners trained on equally-.sized Gaussian clusters; average MSE ratios and confidence intervals are taken over 100 replicates at each level. (A) Fixed dimensional regime, with p = 100, n = 5000, and number of clusters k ranging from 2 to 8. (B) High dimensional regime, with $\gamma = S/n$ ranging from 0 to 0.5, $k = 2$, and $n = 5000$. The theoretical limit in Theorem 4 is shown in red.

$$\lim_{n\to\infty} \frac{\mathbb{E}\left[Var\left[\hat{Y}_E(\mathbf{x}_\star)\right]\right]}{\mathbb{E}\left[Var\left[\hat{Y}_M(\mathbf{x}_\star)\right]\right]} = \frac{S-\frac{1}{2}}{\left[S - \frac{1+2\lambda_1\lambda_2 - \frac{3\lambda_1^2\lambda_2+2\lambda_1\lambda_2}{1+\lambda_1} + \frac{\lambda_1^2\lambda_2+\lambda_1^3\lambda_2}{(1+\lambda_1)^2}}{2+\frac{\lambda_1^2\lambda_2-\lambda_1\lambda_2}{1+\lambda_1}+\lambda_1\lambda_2}\right]} > 1$$

The proof is described in the Appendix, and makes use of Lemmas 8 and 9. Theorem 3 shows us that given assumptions that make the variance calculations comparable, the *Merged* asymptotically produces predictions that are strictly more accurate than the *Ensemble*. Note that in the case that clusters have equal size such that $\lambda_1 = \lambda_2 = 1/2$, this ratio simplifies to $\left(\frac{S-\frac{1}{2}}{S-1}\right)$. Overall, we conclude that when both the *Merged* and *Ensemble* are unbiased, there is no benefit to cluster-specific ensembling when the dimension is fixed.

3.3.3. HIGH DIMENSIONAL ASYMPTOTICS

We now consider the case in which the dimension p grows proportionally with n as $n, S \to \infty$. The following theorem presents the limits of Equations 1a and 1b given this new paradigm.

Theorem 4

We consider a training set with n samples, S covariates, and $k = 2$ Gaussian clusters $\{(\mathbf{x}_i, Y_i) \overset{\text{i.i.d.}}{\sim} \left(\sum_{b=1}^{2} \mathbb{1}\{\mathbf{x}_i \in \textit{Cluster } b\} \times \text{MVN}_S(\boldsymbol{\mu}_\mathbf{b}, \text{I}), \mathbf{x}_i^T\beta\right) : i = 1, \ldots, n\}$, *with* $\beta \in \mathbb{R}_{S\times 1}$.

We denote $\boldsymbol{\mu_b}$ as the cluster-specific means for $b = 1, 2$, and specify that $\boldsymbol{\mu_1} = -\boldsymbol{\mu_2}$. We assume that $n, S \to \infty$ with $\frac{S}{n} = \gamma \in [0, 1/2)$. Then, we can represent the predictions of the Ensemble and the Merged on test point $\mathbf{x}_\star \overset{\text{i.i.d.}}{\sim} \text{MVN}_S(\mathbf{0}, \text{I})$ as $\hat{Y}_E$ and $\hat{Y}_M$, respectively. As $n \to \infty$, the ratio of the mean square errors corresponding to these predictions converges as follows:

$$\lim_{n,S\to\infty} \frac{\mathbb{E}\left[\mathit{Var}\left[\hat{Y}_E(\mathbf{x}_\star)\right]\right]}{\mathbb{E}\left[\mathit{Var}\left[\hat{Y}_M(\mathbf{x}_\star)\right]\right]} = \frac{1-\gamma}{1-2\gamma}$$
$$> 1$$

We conclude once again that the asymptotic variance of the *Merged* is less than that of the *Ensemble* for all $\gamma \in [0, 1/k)$ when $k = 2$. Furthermore, for $\gamma = 0$, indicating fixed dimension S as $n \to \infty$, the ratio in Theorem 4 converges to 1, indicating that the two approaches are asymptotically equal in this case and providing further theoretical evidence to support the empirical findings. Note that the minimal assumptions on $\boldsymbol{\mu_1}$ and $\boldsymbol{\mu_2}$ are different in Theorems 3 and 4, although the intersection of both assumptions is valid in both high dimensional and fixed dimensional scenarios.

The proofs establish that merging outperforms ensembling by comparing two variance expressions derived in Lemma 8. For the *Merged* learner, the expected variance involves the trace of $\left(\mathbf{I} + \sum_b \lambda_b \boldsymbol{\mu}_\mathbf{b} \boldsymbol{\mu}_\mathbf{b}^T\right)^{-1}$, where the cluster means $\boldsymbol{\mu_b}$ enter as rank-1 perturbations that are summed before the matrix is inverted. For the Ensemble, the expected variance is a harmonic mean of traces of $\left(\mathbf{I} + \boldsymbol{\mu}_\mathbf{b} \boldsymbol{\mu}_\mathbf{b}^T\right)^{-1}$, where each cluster's contribution is inverted separately. The Sherman-Morrison calculations in Lemma 9 reveal why these differ: when rank-1 terms combine before inversion, interaction terms arise that further reduce the trace, whereas inverting separately and then averaging doesn't account for these interactions. Theorem 3 confirms the comparison by showing that a particular algebraic quantity $b - 2a$ is negative for all valid cluster proportions, which follows from factoring the expression as $\lambda_1\lambda_2 \times 2(\lambda_1 - 3)(\lambda_1 + 1)$ and observing that $\lambda_1 \in (0, 1)$ makes this product negative. In the high-dimensional setting of Theorem 4, the Inverse Wishart expectation shows that variance scales as $\gamma/(1-\gamma)$ in the aspect ratio γ; since the Ensemble components each operate at effective ratio 2γ while the Merged operates at γ, the ratio of variances becomes $(1-\gamma)/(1-2\gamma)$, which exceeds one for all possible $\gamma \in [0, 1/2)$.

Figure 2 presents the percent change in the average MSE of the *Ensemble* compared to the *Merged* using both the limiting expressions presented in Theorem 4 and results from a simulation study. In the simulation, we set the number of samples per cluster at 400 for $k = 2$, and varied the dimension S incrementally from 0 to 400 to evaluate the performance of these approaches for different values of γ. Clusters were simulated using the `clusterGeneration` package in R, and the coefficients selected to simulate the linear outcome for each cluster were set to be opposites for the first and second clusters. From Figure 1, we observe that the theoretical and empirical results closely mirror one another, and that the performance of the *Ensemble* and the *Merged* are almost identical for values of $\gamma < .8$, after which the *Merged* produces an exponential comparative increase in performance. These results together with the fixed dimension asyptotics presented in Theorem 3 show that for unbiased linear regression, it is overall more advantageous to simply train a single linear regression on the entire dataset than to ensemble regressions trained on clusters.

4. Discussion

Our analysis reveals a central principle underlying cluster-based ensembling: its benefit depends on whether bias or variance dominates the mean squared error (MSE) of the base learner. For Random Forests under measurable, square-integrable targets, the MSE is bias-dominated, and ensembling yields a consistent factor-of-two improvement in risk (Theorem 2). For correctly specified linear regression, where the bias vanishes, variance dominates, and merging strictly outperforms ensembling in both fixed- and high-dimensional regimes (Theorems 3, 4). The same cluster structure can therefore either enhance or degrade accuracy, depending on the bias-variance profile of the learning algorithm.

Although the Random Forest and linear regression analyses adopt different distributional assumptions (uniform clusters for forests and Gaussian clusters for regression), the contrasting outcomes stem not from these distributional choices but from the fundamentally different bias-variance profiles of the two methods under our assumed data-generating processes. Both analyses share key structural elements that enable meaningful comparison: a homogeneous outcome model $f(\mathbf{x})$, distinguishable cluster supports, and weighting schemes that minimize MSE. Their divergence lies in the bias-variance decomposition. Forest predictions suffer bias from the discretization inherent in tree partitioning, which decays more slowly than variance. Ensembling localizes each sub-forest to narrower covariate regions, reducing discretization error and producing the twofold risk improvement noted above. Linear regression, by contrast, is unbiased under correct specification, leaving only variance to be affected by data partitioning. Ensembling thereby reduces the effective sample size and always increases MSE relative to merging, which exploits both the pooled sample and cross-cluster covariance structure. In essence, forests benefit from local specialization, while regression models benefit from global pooling.

Extending these analyses under a unified framework would sharpen their comparability and delineate when cluster-based strategies are beneficial. For our Random Forest analysis, the assumed non-overlapping cluster supports represent a best-case limit for ensembling, as each test point is covered by a single forest. With partial overlap, multiple component learners contribute to each prediction, diminishing localization benefits. In the extreme of full overlap, ensembling and merging coincide, yielding an MSE ratio of one. Empirical results under Gaussian and Laplace mixtures in Figure 1 suggest a smooth interpolation between ratios of one-half (complete separation) and one (full overlap), with observed behavior near $1/\sqrt{2}$ for higher values of k. A common distributional framework, such as expanding the Random Forest analysis to Gaussian clusters via inverse probability transforms mapping Gaussian marginals to uniform coordinates before defining cluster-based partitions, or extending linear regression results to non-Gaussian settings via moment-based arguments, would further clarify how distributional overlap interacts with the bias-variance structure of the estimator.

These theoretical insights yield practical guidance for model selection. When the base learner's MSE is bias-dominated, as in tree-based ensembles, kernel methods with fixed bandwidths, or misspecified parametric models, cluster-based ensembling is likely to yield gains by reducing approximation error even in globally homogeneous outcome settings. Conversely, when variance dominates, as in correctly specified regression or appropriately regularized high-dimensional models, merging typically offers lower risk, with the advantage increasing as dimensionality rises relative to sample size. The effectiveness of ensembling also depends on cluster separation: well-separated clusters permit sharper local fits, whereas high overlap attenuates localization benefits. Assessing

separation through visualization, divergence measures, or cluster classification accuracy can thus inform whether to ensemble or merge.

Several simplifying assumptions bound our analysis. The Random Forest results rely on the Centered Random Forest model proposed by Breiman (2004) which omits bootstrapping and data-dependent splitting; nonetheless, simulations with standard implementations suggest qualitative agreement. The regression analysis assumes equal-sized clusters with known memberships; incorporating estimated or unbalanced clusters would introduce additional variance and potentially alter comparisons. Future work could formalize regimes where bias and variance contribute comparably, such as ridge regression or early-stopped neural networks, and explore optimal weighting under heterogeneous cluster sizes. Finally, developing practical diagnostics to determine whether ensembling or merging is preferable for a given dataset and learner combination would translate our theoretical insights into actionable guidance for applied work.

References

Gérard Biau. Analysis of a random forests model. *Journal of Machine Learning Research*, 13(38): 1063–1095, 2012. URL http://jmlr.org/papers/v13/biau12a.html.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, October 2001. ISSN 0885-6125. doi: 10.1023/A:1010933404324. URL https://doi.org/10.1023/A:1010933404324.

Leo Breiman. Consistency for a simple model of random forests. Technical report, UC Berkeley, 2004.

Ritu Chauhan, Harleen Kaur, and M Afshar Alam. Data clustering method for discovering clusters in spatial cancer databases. *International Journal of Computer Applications*, 10(6):9–14, 2010.

Meghana Deodhar and Joydeep Ghosh. A framework for simultaneous co-clustering and learning from complex data. In *Proceedings of the 13th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '07, page 250–259, New York, NY, USA, 2007. Association for Computing Machinery. ISBN 9781595936097. doi: 10.1145/1281192.1281222. URL https://doi.org/10.1145/1281192.1281222.

Wilson Wen Bin Goh, Wei Wang, and Limsoon Wong. Why batch effects matter in omics data, and how to avoid them. *Trends in Biotechnology*, 35(6):498–507, 2017.

Zoe Guan, Giovanni Parmigiani, and Prasad Patil. Merging versus ensembling in multi-study machine learning: Theoretical insight from random effects. *arXiv*, 2019.

Jason M. Klusowski. Sharp analysis of a simple model for random forests, 2020.

Prasad Patil and Giovanni Parmigiani. Training replicable predictors in multiple studies. *Proceedings of the National Academy of Sciences*, 115(11):2578–2583, 2018. ISSN 0027-8424. doi: 10.1073/pnas.1708283115. URL https://www.pnas.org/content/115/11/2578.

Maya Ramchandran, Prasad Patil, and Giovanni Parmigiani. Tree-weighting for multi-study ensemble learners. *Pacific Symposium on Biocomputing*, 25:451–462, 2020. doi: https://doi.org/10.1101/698779.

Shubhendu Trivedi, Zachary A. Pardos, and Neil T. Heffernan. The utility of clustering in prediction tasks. *CoRR*, abs/1509.06163, 2015. URL http://arxiv.org/abs/1509.06163.

Appendix A. Random Forest proofs

A.1. Proof of Theorem 2

In this section, we provide a a derivation for the limit of the ratio between the bias of the *Ensemble* and the bias of the *Merged* when the number of clusters in the training and testing data is equal to k for $k \geq 2$.

Throughout, we use the following result regarding the leaf node containing test point $\mathbf{x}_\star$.

Remark 5 *Let $a_{nj}(\mathbf{x}_\star, \theta)$ and $b_{nj}(\mathbf{x}_\star, \theta)$ be the left and right endpoints of $A_{nj}(\mathbf{x}_\star, \theta)$, the jth side of the box containing $\mathbf{x}_\star$, for $j = 1, \ldots, S$. For ease of notation, we will henceforth refer to $a_{nj}(\mathbf{x}_\star, \theta)$ as a_{nj} and $b_{nj}(\mathbf{x}_\star, \theta)$ as b_{nj}. $c_{nj}(\mathbf{x}_\star, \theta)$ represents the number of times that the jth coordinate is split, with the total number of splits across all coordinates set to equal $\log_2 c_n$ for some constant $c_n > 2$. For ease of notation, we will suppress the dependencies of a_{nj}, b_{nj}, and c_{nj} on $(\mathbf{x}_\star, \theta)$. We then observe that each endpoint of $A_{nj}(\mathbf{x}_\star, \theta)$ is a randomly stopped binary expansion of $\mathbf{x}_\star^{(j)}$:*

$$a_{nj} \stackrel{D}{=} \frac{1}{k} \sum_{l=1}^{c_{nj}} B_{kj} 2^{-l} + T$$

$$b_{nj} \stackrel{D}{=} \frac{1}{k} \sum_{l=1}^{c_{nj}} B_{kj} 2^{-l} + 2^{-c_{nj}} + T$$

No matter the value of T, the length of the jth side of the box is given by:

$$\lambda(A_{nj}) = b_{nj} - a_{nj} = \frac{1}{k} 2^{-c_{nj}}$$

The division by the number of clusters k reflects the fact that the test point falls within a cluster of width $\frac{1}{k}^S$, so we divide the typical uniform [0,1] binary expansion by the same factor. The measure of the box A_n is therefore equal to

$$\begin{aligned} \lambda(A_n) &= \prod_{j=1}^{S} \lambda(A_{nj}) \\ &= \left(\frac{1}{k}\right)^S 2^{-\lceil \log_2 c_n \rceil} \end{aligned}$$

since by construction, $\sum_j c_{nj} = \lceil \log_2 c_n \rceil$.

The rest of the proof is structured as follows:

(i) We derive the expression for the ratio of the *Ensemble* and *Merged* biases

(ii) We derive expressions for the terms $W_1 W_2'$ and $H_1 H_2'$

(iii) We use (ii) to factor the expressions for the *Ensemble* and *Merged* biases into two terms

(iv) We provide upper and lower bounds for the first term of the factorization

(v) We use these bounds to bound the overall limiting ratio

(vi We simplify the second term of the factorization

(vii) We combine all previous steps and take limits to complete the proof

Proof

(i) The leading term of the squared bias for the *Ensemble* can be represented as

$$\begin{aligned} & n(n-1)\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\mathbb{E}_\theta\left[W_1\right]\left(f(\mathbf{x}_1)-f(\mathbf{x}_\star)\right)\mathbb{E}_\theta\left[W_2\right]\left(f(\mathbf{x}_2)-f(\mathbf{x}_\star)\right)\right] \\ = \; & n(n-1)\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\mathbb{E}_\theta\left[W_1\right]\mathbb{E}_\theta\left[W_2\right]\left(f(\mathbf{x}_1)-f(\mathbf{x}_\star)\right)\left(f(\mathbf{x}_2)-f(\mathbf{x}_\star)\right)\right] \\ = \; & n(n-1)\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta,\theta'}\left[W_1W_2^{'}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))(f(\mathbf{x}_2)-f(\mathbf{x}_\star))\right] \end{aligned}$$

where $W_i(\mathbf{x}_\star,\theta)=\sum_{b=1}^k W_{ib}(\mathbf{x}_\star,\theta)$, as defined in Lemma 1 (i) and $\theta^{'}$ is an independent copy of θ; thus, for any quantity indexed with the $'$ notation, we replace $A_n(\mathbf{x}_\star,\theta)$ with $A_n(\mathbf{x}_\star,\theta^{'})$.

The leading term of the squared bias for the *Merged* is equal to

$$\begin{aligned} & n(n-1)\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\mathbb{E}_\theta\left[H_1\right]\left(f(\mathbf{x}_1)-f(\mathbf{x}_\star)\right)\mathbb{E}_\theta\left[H_2\right]\left(f(\mathbf{x}_2)-f(\mathbf{x}_\star)\right)\right] \\ = \; & n(n-1)\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta,\theta'}\left[H_1H_2^{'}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))(f(\mathbf{x}_2)-f(\mathbf{x}_\star))\right] \end{aligned}$$

where H_1 and $H_2^{'}$ are defined in Lemma 1 (ii).

Therefore, the limiting ratio of the *Ensemble* bias vs. the *Merged* bias is

$$\lim_{n\to\infty}\sqrt{\frac{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta,\theta'}\left[W_1W_2^{'}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))(f(\mathbf{x}_2)-f(\mathbf{x}_\star))\right]}{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta,\theta'}\left[H_1H_2^{'}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))(f(\mathbf{x}_2)-f(\mathbf{x}_\star))\right]}}$$

For the rest of the proof, we focus on evaluating the ratio within the square root, and applying the square root function as the last step.

(ii) To simplify this further, we expand the quantities $W_1W_2^{'}$ and $H_1H_2^{'}$.

$$\begin{aligned} & W_1W_2^{'} \\ & = \mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x_2}\in A_n'\}} \\ & \times\left[\sum_{b=1}^k\frac{\mathbb{1}_{\{\mathbf{x}_1\in[\frac{b-1}{k},\frac{b}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[\frac{b-1}{k},\frac{b}{k}]^S\}}}{N_b}\mathbb{1}_{\{\epsilon_{nb}\}}\right]\times\left[\sum_{b=1}^k\frac{\mathbb{1}_{\{\mathbf{x}_1\in[\frac{b-1}{k},\frac{b}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[\frac{b-1}{k},\frac{b}{k}]^S\}}}{N_b^{'}}\mathbb{1}_{\{\epsilon_{nb}'\}}\right] \\ & = \mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x_2}\in A_n'\}} \\ & \times\left[\sum_{b=1}^k\frac{\mathbb{1}_{\{\mathbf{x}_1\in[\frac{b-1}{k},\frac{b}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_2\in[\frac{b-1}{k},\frac{b}{k}]^S\}}[\mathbb{1}_{\{\mathbf{x}_\star\in[\frac{b-1}{k},\frac{b}{k}]^S\}}]^2\mathbb{1}_{\{\epsilon_{nb}\}}\mathbb{1}_{\{\epsilon_{nb}'\}}}{N_bN_b'}\right] \end{aligned}$$

The cross terms of the product are equal to 0, since $\mathbf{x_1}, \mathbf{x_1},$ and $\mathbf{x}_\star$ have to fall in the same cluster for any term to be non-zero.

Similarly,

$$H_1 H_2' = \mathbb{1}_{\{\mathbf{x_1} \in A_n\}} \mathbb{1}_{\{\mathbf{x_2} \in A_n'\}} \times \frac{\mathbb{1}_{\{\epsilon_n\}}}{N_n} \times \frac{\mathbb{1}_{\{\epsilon_n'\}}}{N_n'}$$

(iii) We now simplify these quantities using the following definitions and assumptions: Define

$$T_b = \sum_{i \geq 3} \mathbb{1}_{\{\mathbf{x}_i \in A_n\}} \mathbb{1}_{\{\mathbf{x}_i \in [\frac{b-1}{k}, \frac{b}{k}]^S\}} \mathbb{1}_{\{\mathbf{x}_\star \in [\frac{b-1}{k}, \frac{b}{k}]^S\}}$$

for $b = 1, \ldots, k$ and let $T_b^{'}$ be the equivalent expressions based on $\theta^{'}$. We make the assumption that the quantities $\mathbb{1}_{\{\epsilon_{nb}\}}$, $b = 1, \ldots, k$, as well as $\mathbb{1}_{\{\epsilon_n\}}$, are all non-zero; that is, that there is always at least one training point that falls within $A_n(\mathbf{x}_\star, \theta)$, within the box (leaf node) containing the test point, regardless of the test point's true cluster assignment. Without loss of generality, we can therefore order the dataset so that one of $\mathbf{x}_1$, $\mathbf{x}_2$ falls within $A_n(\mathbf{x}_\star, \theta)$ and the other falls in another cluster. For any test point, we can identify at least k training data points that belong to each distinct cluster out of which one is guaranteed to fall within the same leaf node as the test point. There are therefore a minimum of $k/2$ different arrangements of the training set that meet this criteria for all $b = 1, \ldots, k$.

For test points belonging to cluster b, we denote the event that this assumption is met by $d_b = \mathbb{1}_{\left\{\epsilon_{nb} > 0, \sum_{i=1}^2 \mathbb{1}_{\{\mathbf{x}_i \in A_n\}} = 1, \sum_{i=1}^2 \mathbb{1}\left\{\mathbf{x}_i \in \left[\frac{b-1}{k}, \frac{b}{k}\right]^S\right\} = 1\right\}}$. For any test point belonging to an arbitrary cluster, we denote the event that this assumption is met by $d_n = \mathbb{1}_{\{\epsilon_n > 0, \sum_{i=1}^2 \mathbb{1}\{\mathbf{x}_i \in A_n\} = 1\}}$. This event is in fact exactly equivalent to our overall assumption, so by definition, $P(\mathcal{D}_n) = 1$. Therefore,

$$\begin{aligned}
\mathbb{E}\left[\frac{\mathbb{1}_{\{\epsilon_{nb}\}} \mathbb{1}_{\{\epsilon_{nb}'\}}}{N_b N_b'} \middle| \mathbf{x}_\star\right] &= \mathbb{E}\left[\frac{1}{(1+T_b)(1+T_b')} \middle| \mathbf{x}_\star, d_b\right] \\
&= \frac{\mathbb{E}\left[\frac{1}{(1+T_b)(1+T_b')} \middle| \mathbf{x}_\star\right]}{P(d_b)} \\
&= \frac{k}{2} \mathbb{E}\left[\frac{1}{(1+T_b)(1+T_b')} \middle| \mathbf{x}_\star\right]
\end{aligned}$$

for $b = 1, \ldots, k$. For the *Merged* learner, we can similarly define

$$U = \sum_{i \geq 3} \mathbb{1}_{\{\mathbf{x}_i \in A_n\}}$$

and

$$\begin{aligned}
\mathbb{E}\left[\frac{\mathbb{1}_{\{\epsilon_n\}} \mathbb{1}_{\{\epsilon_n'\}}}{N_n N_n'} \middle| \mathbf{x}_\star\right] &= \mathbb{E}\left[\frac{1}{(1+U)(1+U')} \middle| \mathbf{x}_\star, d_n\right] \\
&= \mathbb{E}\left[\frac{1}{(1+U)(1+U')} \middle| \mathbf{x}_\star\right]
\end{aligned}$$

Given these definitions, we can further simplify the ratio of the squared biases:

$$\frac{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta,\theta'}\left[W_1 W_2^{'}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))(f(\mathbf{x}_2)-f(\mathbf{x}_\star))\right]}{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta,\theta'}\left[H_1 H_2^{'}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))(f(\mathbf{x}_2)-f(\mathbf{x}_\star))\right]}$$

(iiia) The numerator (corresponding to the leading term of the squared bias of the *Ensemble*) simplifies to:

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta,\theta'}\left[W_1 W_2^{'}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))(f(\mathbf{x}_2)-f(\mathbf{x}_\star))\right] \\
=\ &\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta,\theta'}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x_2}\in A_n'\}}\times\left[\sum_{b=1}^{k}\frac{\mathbb{1}_{\{\mathbf{x}_1\in[\frac{b-1}{k},\frac{b}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_2\in[\frac{b-1}{k},\frac{b}{k}]^S\}}\left[\mathbb{1}_{\{\mathbf{x}_\star\in[\frac{b-1}{k},\frac{b}{k}]^S\}}\right]^2\mathbb{1}_{\{\epsilon_{nb}\}}\mathbb{1}_{\{\epsilon_{nb}'\}}}{N_b N_b'}\right]\right] \\
=\ &\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta,\theta'}\left[\mathbb{E}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x_2}\in A_n'\}}\times k\left[\frac{\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_2\in[0,\frac{1}{k}]^S\}}\left[\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}\right]^2\mathbb{1}_{\{\epsilon_{n1}\}}\mathbb{1}_{\{\epsilon_{n1}'\}}}{N_1 N_1'}\right]\middle|\mathbf{x}_\star,\theta,\theta^{'}\right]\right] \\
=\ &k\times\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\mathbb{E}\left[\frac{1}{(1+T_1)(1+T_1')}\middle|\mathbf{x}_\star,\theta,\theta^{'},d_b\right]\right. \\
&\times\mathbb{E}_{\mathbf{x}_1,\mathbf{x}_2}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x_2}\in A_n'\}}\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_2\in[0,\frac{1}{k}]^S\}}\left[\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}\right]^2\right. \\
&\left.\left.\times(f(\mathbf{x}_1)-f(\mathbf{x}_\star))(f(\mathbf{x}_2)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star,\theta,\theta^{'}\right]\right] \\
=\ &\frac{k^2}{2}\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\mathbb{E}\left[\frac{1}{(1+T_1)(1+T_1')}\middle|\mathbf{x}_\star,\theta,\theta^{'}\right]\right. \\
&\left.\times\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]
\end{aligned}$$

Various independence relations are used to simplify the expression: $\mathbf{x}_1 \perp \mathbf{x}_2$, $\theta \perp \theta^{'}$, and $T_1 \perp \mathbf{x}_1, \mathbf{x}_2$.

(iiib) The denominator (corresponding to the leading term of the squared bias of the *Merged*) simplifies to:

$$
\begin{aligned}
&\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta,\theta'}\left[H_1 H_2^{'}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))(f(\mathbf{x}_2)-f(\mathbf{x}_\star))\right] \\
&\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta,\theta'}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x_2}\in A_n^{'}\}}\left(\frac{1}{1+U}\right)\left(\frac{1}{1+U'}\right)(f(\mathbf{x}_1)-f(\mathbf{x}_\star))(f(\mathbf{x}_2)-f(\mathbf{x}_\star))\right] \\
&\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta}\left[\left(\frac{1}{1+U}\right)\left(\frac{1}{1+U'}\right)\left(\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\right)^2\right] \\
=\ &\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\mathbb{E}\left[\frac{1}{(1+U)(1+U')}|\mathbf{x}_\star,\theta,\theta^{'}\right]\times\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))|\mathbf{x}_\star\right]\right)^2\right]
\end{aligned}
$$

(iv) We now focus on calculating $\mathbb{E}\left[\frac{1}{(1+T_1)(1+T_1^{'})}|\mathbf{x}_\star,\theta,\theta^{'}\right]$ and $\mathbb{E}\left[\frac{1}{(1+U)(1+U')}|\mathbf{x}_\star,\theta,\theta^{'}\right]$. To upper bound these quantities, we use the fact that for $Z\sim$ Binomial(m,p), $\mathbb{E}\left[\left(\frac{1}{1+Z}\right)^2\right]$ $\leq \frac{1}{(m+1)(m+2)p^2}$ along with the Cauchy-Schwarz inequality. To lower bound these quantities, we use the fact that for $Z\sim$ Binomial(m,p), one has that $\mathbb{E}\left[\frac{1}{1+Z}\right]=\frac{1}{(m+1)p}\left[1-(1-p)^{m+1}\right]$ along with the definition of covariance, which tells us that $\text{Cov}(X,Y)=E[XY]-E[X]E[Y]\Rightarrow$ $E[X]E[Y]\leq E[XY]$ for positive $\text{Cov}(X,Y)$. To move forward, we need to calculate the binomial probabilities associated with T_1 and U.

$T_1\sim$ Binomial$(n-2,p_e)$, where

$$
\begin{aligned}
p_e &= \mathbb{E}\left[\mathbb{1}_{\{\mathbf{x}_i\in A_n\}}\mathbb{1}_{\{\mathbf{x}_i\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}\right] \\
&= \mathbb{E}\left[\mathbb{1}_{\{\mathbf{x}_i\in A_n\}}\middle|\mathbb{1}_{\{\mathbf{x}_i\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}\right]\times\mathbb{E}\left[\mathbb{1}_{\{\mathbf{x}_i\in[0,\frac{1}{k}]^S\}}\right]\times\mathbb{E}\left[\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}\right] \\
&= \frac{1}{k^2}\prod_{j=1}^{S}\frac{b_{nj}-a_{nj}}{\frac{1}{k}} \\
&= \frac{1}{k^2}2^{-\lceil\log_2 c_n\rceil}
\end{aligned}
$$

$U \sim \text{Binomial}(n-2, p_m)$ where

$$\begin{aligned}
p_m &= E_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1} \in A_n\}}\right] \\
&= \prod_{j=1}^{S} k \times P\left(\mathbb{1}_{\{\mathbf{x}_1^{(j)} \in A_{nj}\}} \middle| \mathbf{x}_1^{(j)} \in \left[0, \frac{1}{k}\right]\right) \mathbf{P}\left(\mathbf{x}_1^{(j)} \in \left[0, \frac{1}{k}\right]\right) \\
&= \prod_{j=1}^{S} P\left(\mathbb{1}_{\{\mathbf{x}_1^{(j)} \in A_{nj}\}} \middle| \mathbf{x}_1^{(j)} \in \left[0, \frac{1}{k}\right], \mathbf{x}_\star^{(j)} \in \left[0, \frac{1}{k}\right]\right) \mathbf{P}\left(\mathbf{x}_\star^{(j)} \in \left[0, \frac{1}{k}\right]\right) \\
&= \prod_{j=1}^{S} \frac{b_{nj} - a_{nj}}{\frac{1}{k}} \times \frac{1}{k} \\
&= \frac{1}{k} 2^{-\lceil \log_2 c_n \rceil}
\end{aligned}$$

Now, we can state that by the definition of covariance,

$$\begin{aligned}
\mathbb{E}\left[\frac{1}{(1+T_1)(1+T_1')} \middle| \mathbf{x}_\star, \theta, \theta'\right] &\geq \mathbb{E}\left[\frac{1}{(1+T_1)} \middle| \mathbf{x}_\star, \theta, \theta'\right] \mathbb{E}\left[\frac{1}{(1+T_1')} \middle| \mathbf{x}_\star, \theta, \theta'\right] \\
&= \left(\mathbb{E}\left[\frac{1}{(1+T_1)} \middle| \mathbf{x}_\star, \theta, \theta'\right]\right)^2 \\
&= \frac{k^4 4^{\lceil \log_2 c_n \rceil}}{(n-1)^2} \left[1 - \left(\frac{1}{k^2} 2^{-\lceil \log_2 c_n \rceil}\right)^{n-1}\right]^2
\end{aligned}$$

By the Cauchy-Schwarz inequality,

$$\begin{aligned}
&\mathbb{E}\left[\frac{1}{(1+T_1)(1+T_1')} \middle| \mathbf{x}_\star, \theta, \theta'\right] \\
&\leq \sqrt{\mathbb{E}\left[\left(\frac{1}{1+T_1}\right)^2 \middle| \mathbf{x}_\star, \theta, \theta'\right]} \sqrt{\mathbb{E}\left[\left(\frac{1}{1+T_1'}\right)^2 \middle| \mathbf{x}_\star, \theta, \theta'\right]} \\
&= \left(\frac{k^2 2^{\lceil \log_2 c_n \rceil}}{\sqrt{n(n-1)}}\right)^2 \\
&= \frac{k^4 4^{\lceil \log_2 c_n \rceil}}{n(n-1)}
\end{aligned}$$

Putting it all together,

$$\frac{k^4 4^{\lceil \log_2 c_n \rceil}}{(n-1)^2} \left[1 - \left(\frac{1}{k^2} 2^{-\lceil \log_2 c_n \rceil}\right)^{n-1}\right]^2 \leq \mathbb{E}\left[\frac{1}{(1+T_1)(1+T_1')} \middle| \mathbf{x}_\star, \theta, \theta'\right] \leq \frac{k^4 4^{\lceil \log_2 c_n \rceil}}{n(n-1)}$$

Now, we can do a similar calculation for $\mathbb{E}\left[\frac{1}{(1+U)(1+U')} | \mathbf{x}_\star, \theta, \theta'\right]$, yielding:

$$\frac{k^2 4^{\lceil \log_2 c_n \rceil}}{(n-1)^2}\left[1-\left(\frac{1}{k}2^{-\lceil \log_2 c_n \rceil}\right)^{n-1}\right]^2 \leq \mathbb{E}\left[\frac{1}{(1+U)(1+U')}\middle|\mathbf{x}_\star,\theta,\theta'\right] \leq \frac{k^2 4^{\lceil \log_2 c_n \rceil}}{n(n-1)}$$

(v) To get an upper bound on our ratio simplified in part (iii), we can take the max of the numerator and the min of the denominator:

$$\begin{aligned}
&\max \frac{\frac{k^2}{2}\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\begin{array}{c}\mathbb{E}\left[\frac{1}{(1+T_1)(1+T_1')}|\mathbf{x}_\star,\theta,\theta'\right] \\ \times\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\end{array}\right]}{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\mathbb{E}\left[\frac{1}{(1+U)(1+U')}|\mathbf{x}_\star,\theta,\theta'\right]\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]} \\
&= \frac{\frac{k^2}{2}\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\frac{k^4 4^{\lceil \log_2 c_n \rceil}}{n(n-1)}\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]}{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\frac{k^2 4^{\lceil \log_2 c_n \rceil}}{(n-1)^2}\left[1-\left(\frac{1}{k}2^{-\lceil \log_2 c_n \rceil}\right)^{n-1}\right]^2\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]} \\
&= \left(\frac{k^4}{2}\right)\left(\frac{n-1}{n}\right)\left(\frac{1}{\left[1-\left(\frac{1}{k}2^{-\lceil \log_2 c_n \rceil}\right)^{n-1}\right]^2}\right) \\
&\times \frac{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]}{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]}
\end{aligned}$$

We find the lower bound of the ratio by taking the min of the numerator and max of the denominator, yielding:

$$\min \frac{\frac{k^2}{2}\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\begin{array}{c}\mathbb{E}\left[\frac{1}{(1+T_1)(1+T_1')}\middle|\mathbf{x}_\star,\theta,\theta'\right] \\ \times\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\end{array}\right]}{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\mathbb{E}\left[\frac{1}{(1+U)(1+U')}\middle|\mathbf{x}_\star,\theta,\theta'\right]\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]}$$

$$= \left(\frac{k^4}{2}\right)\left(\frac{n}{n-1}\right)\left[1-\left(\frac{1}{k^2}2^{-\lceil\log_2 c_n\rceil}\right)^{n-1}\right]^2$$

$$\times \frac{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]}{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]}$$

(vi) We now simplify the quantity

$$\frac{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]}{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]}$$

We begin with the numerator, evaluating the expression within the inner expectation:

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}f(\mathbf{x}_1)-f(\mathbf{x}_\star)\middle|\mathbf{x}_\star\right] \\
=\ &\mathbb{E}_{\mathbf{x}_1}\left[f(\mathbf{x}_1)-f(\mathbf{x}_\star)\middle|\mathbf{x_1}\in A_n,\mathbf{x}_1\in\left[0,\frac{1}{k}\right]^S,\mathbf{x}_\star\in\left[0,\frac{1}{k}\right]^S,\mathbf{x}_\star\right] \\
&\times\mathbb{E}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}|\mathbf{x}_1\in\left[0,\frac{1}{k}\right]^S,\mathbf{x}_\star\in\left[0,\frac{1}{k}\right]^S,\mathbf{x}_\star\right]\times\mathbb{E}\left[\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\right]\times\mathbb{E}\left[\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}\right] \\
=\ &\left(\frac{\lambda(A_n)}{k}\right)\mathbb{E}_{\mathbf{x}_1}\left[f(\mathbf{x}_1)-f(\mathbf{x}_\star)\middle|\mathbf{x_1}\in A_n,\mathbf{x}_\star\right]
\end{aligned}$$

Moving onto the denominator, we again evaluate the expression within the inner expectation:

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\\
=\;&\mathbb{E}_{\mathbf{x}_1}\left[f(\mathbf{x}_1)-f(\mathbf{x}_\star)\middle|\mathbf{x_1}\in A_n,\mathbf{x}_\star\right]\times\mathbb{E}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\middle|\mathbf{x}_\star\right]\\
=\;&\mathbb{E}_{\mathbf{x}_1}\left[f(\mathbf{x}_1)-f(\mathbf{x}_\star)\middle|\mathbf{x_1}\in A_n,\mathbf{x}_\star\right]\frac{\lambda(A_n)}{\frac{1}{k}}
\end{aligned}$$

Therefore,

$$\begin{aligned}
&\frac{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]}{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))|\mathbf{x}_\star\right]\right)^2\right]}\\
=\;&\frac{\left(\frac{\lambda(A_n)}{k}\right)^2\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\left(\mathbb{E}_{\mathbf{x}_1}\left[f(\mathbf{x}_1)-f(\mathbf{x}_\star)\middle|\mathbf{x_1}\in A_n,\mathbf{x}_\star\right]\right)^2\right]}{(k\lambda(A_n))^2\,\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\left(\mathbb{E}_{\mathbf{x}_1}\left[f(\mathbf{x}_1)-f(\mathbf{x}_\star)\middle|\mathbf{x_1}\in A_n,\mathbf{x}_\star\right]\right)^2\right]}\\
=\;&\frac{1}{k^4}
\end{aligned}$$

Note that we do not need to make any assumptions on the form that $f(x)$ takes, since the expressions involving the outcome model cancel between numerator and denominator.

(vii) Putting everything together from parts (v) and (vi),

$$\begin{aligned}
&\frac{\frac{k^2}{2}\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\begin{array}{c}\mathbb{E}\left[\frac{1}{(1+T_1)(1+T_1')}|\mathbf{x}_\star,\theta,\theta'\right]\\ \times\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\end{array}\right]}{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\mathbb{E}\left[\frac{1}{(1+U)(1+U')}|\mathbf{x}_\star,\theta,\theta'\right]\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]}\\
\geq\;&\left(\frac{k^4}{2}\right)\left(\frac{n}{n-1}\right)\left[1-\left(\frac{1}{k^2}2^{-\lceil\log_2 c_n\rceil}\right)^{n-1}\right]^2\times\frac{1}{k^4}\\
=\;&\left(\frac{n}{n-1}\right)\frac{1}{2}\left[1-\left(\frac{1}{k^2}2^{-\lceil\log_2 c_n\rceil}\right)^{n-1}\right]^2
\end{aligned}$$

and

$$
\begin{aligned}
&\frac{\frac{k^2}{2}\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\begin{array}{c}\mathbb{E}\left[\frac{1}{(1+T_1)(1+T_1')}|\mathbf{x}_\star,\theta,\theta'\right]\\ \times\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x}_1\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\end{array}\right]}{\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\mathbb{E}\left[\frac{1}{(1+U)(1+U')}|\mathbf{x}_\star,\theta,\theta'\right]\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}(f(\mathbf{x}_1)-f(\mathbf{x}_\star))\middle|\mathbf{x}_\star\right]\right)^2\right]}\\
&\leq \left(\frac{k^4}{2}\right)\left(\frac{n-1}{n}\right)\times\frac{1}{k^4}\left(\frac{1}{\left[1-\left(\frac{1}{k}2^{-\lceil\log_2 c_n\rceil}\right)^{n-1}\right]^2}\right)\\
&= \left(\frac{n}{n-1}\right)\frac{1}{2}\left(\frac{1}{\left[1-\left(\frac{1}{k}2^{-\lceil\log_2 c_n\rceil}\right)^{n-1}\right]^2}\right)
\end{aligned}
$$

We now calculate the ratio of the *Ensemble* bias vs. the *Merged* bias (not the squared ratio) and take limits of both directions as $n\to\infty$, noting that both left and right endpoints converge to the same value.

$$
\begin{aligned}
\lim_{n\to\infty}\frac{\textit{Ensemble bias}}{\textit{Merged bias}} &\in \left[\begin{array}{c}\lim_{n\to\infty}\left(\frac{n}{n-1}\right)\frac{1}{2}\left[1-\left(\frac{1}{k^2}2^{-\lceil\log_2 c_n\rceil}\right)^{n-1}\right]^2,\\ \lim_{n\to\infty}\left(\frac{n}{n-1}\right)\frac{1}{2}\left(\frac{1}{\left[1-\left(\frac{1}{k}2^{-\lceil\log_2 c_n\rceil}\right)^{n-1}\right]^2}\right)\end{array}\right]\\
&\to\left[\frac{1}{2},\frac{1}{2}\right]\\
&\Rightarrow\lim_{n\to\infty}\frac{\textit{Ensemble bias}}{\textit{Merged bias}}\to\frac{1}{2}
\end{aligned}
$$

for any $k\geq 2$

■

Remark 6 *Convergence of bias terms*

(i) From the proof of Theorem 1, we can express the squared bias of the Ensemble as:

$$
\begin{aligned}
&n(n-1)\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta,\theta'}\left[W_1W_2'(f(\mathbf{x}_1)-f(\mathbf{x}_\star))(f(\mathbf{x}_2)-f(\mathbf{x}_\star))\right]\\
\to\ &\frac{k^{4-2S}}{2}\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\left(\mathbb{E}_{\mathbf{x}_1}\left[f(\mathbf{x}_1)-f(\mathbf{x}_\star)\middle|\mathbf{x_1}\in A_n,\mathbf{x}_\star\right]\right)^2\right.\\
=\ &\Omega\left(\frac{1}{n}\right)
\end{aligned}
$$

The convergence of this term depends on the function $f(x)$, distributions of $\mathbf{x}_1$ and $\mathbf{x}_\star$, number of non-sparse variables S, the number of clusters k, and the total number of splits c_n (which influences the size of the box containing $\mathbf{x}_\star$). Assuming that $\frac{k}{n}$, $\frac{c_n}{n}$, and $\frac{S}{n} \to 0$ as $n \to \infty$, the squared bias converges at a rate greater than $O\left(\frac{1}{n}\right)$, which we represent as $\Omega\left(\frac{1}{n}\right)$

(ii) We can express the squared bias of the Merged as:

$$\begin{aligned} & n(n-1)\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n,\theta,\theta'}\left[H_1 H_2'(f(\mathbf{x}_1)-f(\mathbf{x}_\star))(f(\mathbf{x}_2)-f(\mathbf{x}_\star))\right] \\ \to\ & k^{4-2S}\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\left(\mathbb{E}_{\mathbf{x}_1}\left[f(\mathbf{x}_1)-f(\mathbf{x}_\star)\middle|\mathbf{x_1}\in A_n,\mathbf{x}_\star\right]\right)^2\right] \\ =\ & \Omega\left(\frac{1}{n}\right) \end{aligned}$$

using similar arguments as above.

A.2. Lemma 7

We now show that the variance of the *Ensemble* and *Merged* asymptotically converge to zero. We assume that $\text{Var}[f(\mathbf{x})|\mathbf{x}] \leq \sigma^2$ for some $\sigma > 0$.

Lemma 7

(i) Using [Biau 2012, Proposition 2], we can express the variance of the Ensemble as

$$\begin{aligned} \mathbb{E}[\hat{Y}_E(\mathbf{x}_\star)-\mathbb{E}[\hat{Y}_E(\mathbf{x}_\star)|\mathbf{x}_\star]]^2 &= n\sigma^2\mathbb{E}\left[\mathbb{E}_{\theta,\theta'}\left[W_1 W_2'\right]\right] \\ &= n\sigma^2\frac{k^2}{2}\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\mathbb{E}\left[\frac{1}{(1+T_1)(1+T_1')}\middle|\mathbf{x}_\star,\theta,\theta'\right]\right. \\ &\quad \left.\times\left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\mathbb{1}_{\{\mathbf{x_1}\in[0,\frac{1}{k}]^S\}}\mathbb{1}_{\{\mathbf{x}_\star\in[0,\frac{1}{k}]^S\}}\middle|\mathbf{x}_\star\right]\right)^2\right] \\ &= n\sigma^2\frac{k^2}{2}\frac{k^4 4^{\lceil\log_2 c_n\rceil}}{n(n-1)}\left(\frac{\lambda(A_n)}{\frac{1}{k}}\left(\frac{1}{k}\right)^2\right)^2 \\ &= \frac{\sigma^2 k^{(4-2S)}}{2(n-1)} \\ &= O\left(\frac{1}{n}\right) \end{aligned}$$

using calculations from the proof of Theorem 1, step (iiia), (iv), and (vi) and finite assumptions on S and k outlined in Remark 6. Note that we do NOT need a finite assumption on the number of total splits c_n for the variance to be $O(1/n)$.

(ii) Similarly, we can express the variance of the Merged as

$$\begin{aligned}
&\mathbb{E}[\hat{Y}_M(\mathbf{x}_\star) - \mathbb{E}[\hat{Y}_M(\mathbf{x}_\star)|\mathbf{x}_\star]]^2 \\
&= n\sigma^2\mathbb{E}\left[\mathbb{E}_{\theta,\theta'}\left[H_1 H_2^{'}\right]\right] \\
&= n\sigma^2\mathbb{E}_{\mathbf{x}_\star,\mathcal{D}_n}\left[\mathbb{E}\left[\frac{1}{(1+U)(1+U')}\middle|\mathbf{x}_\star,\theta,\theta^{'}\right] \times \left(\mathbb{E}_{\mathbf{x}_1}\left[\mathbb{1}_{\{\mathbf{x_1}\in A_n\}}\middle|\mathbf{x}_\star\right]\right)^2\right] \\
&= n\sigma^2\frac{k^2 4^{\lceil \log_2 c_n\rceil}}{n(n-1)}\left(\frac{\lambda(A_n)}{\frac{1}{k}}\right)^2 \\
&= \frac{\sigma^2 k^{(3-2S)}}{n-1} \\
&= O\left(\frac{1}{n}\right)
\end{aligned}$$

using calculations from the proof of Theorem 2, step (iiib), (iv), and (vi).

Therefore, the variance terms converge to 0 as $n \to \infty$ and the squared bias asymptotically dominates the MSE.

Appendix B. Linear Least Squares Proofs

B.1. Lemma 8

Lemma 8 *Assume that n_b is the number of samples in the bth cluster and that $n = \sum_{t=1}^{k} n_b$ for $b = 1, 2$. Then, taking expectations of Equations 1a and 1b with respect to all random quantities, we obtain the following expressions:*

(i)

$$\lim_{n\to\infty} \frac{\mathbb{E}\left[Var\left[\hat{Y}_E(\mathbf{x}_\star)\right]\right]}{d_1} = 1$$

(ii)

$$\lim_{n\to\infty} \frac{\mathbb{E}\left[Var\left[\hat{Y}_M(\mathbf{x}_\star)\right]\right]}{d_2} = 1$$

where

$$\begin{aligned}
d_1 &= \left[\sum_{b=1}^{k}\left(\frac{1}{n_b}\left[(S-1) + \frac{1}{1+\|\boldsymbol{\mu}_\mathbf{b}\|^2}\right]\right)^{-1}\right]^{-1} \\
d_2 &= \frac{1}{n} Trace\left(\mathrm{I} + \sum_{b=1}^{k}\frac{n_b}{n}\boldsymbol{\mu}_\mathbf{b}\boldsymbol{\mu}_\mathbf{b}^T\right)^{-1}
\end{aligned}$$

Proof

(i) We begin by considering the term $\mathbf{x}_\star^T(\mathbb{X}_b^T\mathbb{X}_b)^{-1}\mathbf{x}_\star$, and taking expectations with respect to the random quantities:

$$\mathbf{x}_\star^T(\mathbb{X}_b^T\mathbb{X}_b)^{-1}\mathbf{x}_\star = \frac{1}{n_b}\mathbf{x}_\star^T\left(\frac{\mathbb{X}_b^T\mathbb{X}_b}{n^t}\right)^{-1}\mathbf{x}_\star$$

Recall that n_b denotes the number of observations in the bth training cluster, and $\mathbb{X}_b = [\mathbf{x_1}, ..., \mathbf{x_{n_b}}]^T$, where $\mathbf{x_i}$ is a $S \times 1$ vector for $i = 1, ..., n_b$. Then,

$$\begin{aligned}\frac{\mathbb{X}_b^T\mathbb{X}_b}{n^t} &= \frac{1}{n_b}\sum_{i=1}^{n_b}\mathbf{x_i}^T\mathbf{x_i} \\ &\xrightarrow{p} \mathbb{E}[\mathbf{x_1}^T\mathbf{x_1}] \\ &= \text{Var}[\mathbf{x_1}] + E[\mathbf{x_1}]^2 \\ &= \mathrm{I} + \boldsymbol{\mu_b}\boldsymbol{\mu_b}^T\end{aligned}$$

Therefore,

$$\frac{1}{n_b}\mathbb{E}_{\mathbb{X}_b}\left[\mathbf{x}_\star^T\left(\frac{\mathbb{X}_b^T\mathbb{X}_b}{n^t}\right)^{-1}\mathbf{x}_\star\right] = \frac{1}{n_b}\mathbf{x}_\star^T(\mathrm{I} + \boldsymbol{\mu_b}\boldsymbol{\mu_b}^T)^{-1}\mathbf{x}_\star$$

We now take the expectation of this expression with respect to the random quantity $\mathbf{x}_\star$:

$$\begin{aligned}\frac{1}{n_b}\mathbb{E}_{\mathbf{x}_\star}\left[\mathbf{x}_\star^T(\mathrm{I} + \boldsymbol{\mu_b}\boldsymbol{\mu_b}^T)^{-1}\mathbf{x}_\star\right] &= \frac{1}{n_b}\text{Trace}\left[(\mathrm{I} + \boldsymbol{\mu_b}\boldsymbol{\mu_b}^T)^{-1}\right] \\ &= \frac{1}{n_b}\left[(S-1) + \frac{1}{1+\|\boldsymbol{\mu_b}\|^2}\right]\end{aligned}$$

This shows us that:

$$\begin{aligned}\mathbb{E}\left[\mathbf{x}_\star^T(\mathbb{X}_b^T\mathbb{X}_b)^{-1}\mathbf{x}_\star\right] &= \mathbb{E}_{\mathbf{x}_\star}\left[\mathbb{E}_{\mathbb{X}_b}\left[\mathbf{x}_\star^T(\mathbb{X}_b^T\mathbb{X}_b)^{-1}\mathbf{x}_\star\right]\right] \\ &= \frac{1}{n_b}\left[(S-1) + \frac{1}{1+\|\boldsymbol{\mu_b}\|^2}\right]\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbb{E}\left[\text{Var}\left[\hat{Y}_E(\mathbf{x}_\star)\right]\right] &= \mathbb{E}\left[\sum_{b=1}^{k}\left[\mathbf{x}_\star^T(\mathbb{X}_b^T\mathbb{X}_b)^{-1}\mathbf{x}_\star\right]^{-1}\right]^{-1} \\ &= \left[\sum_{b=1}^{k}\left(\frac{1}{n_b}\left[(S-1) + \frac{1}{1+\|\boldsymbol{\mu_b}\|^2}\right]\right)^{-1}\right]^{-1}\end{aligned}$$

(ii) We begin by taking expectations with respect to $\mathbf{x}_\star$

$$\begin{aligned}\mathbb{E}_{\mathbf{x}_\star}\left[\text{Var}(\hat{Y}_m)\right] &= \mathbb{E}_{\mathbf{x}_\star}\left[\mathbf{x}_\star^T\left(\sum_{b=1}^{k}\mathbb{X}_b^T\mathbb{X}_b\right)^{-1}\mathbf{x}_\star\right] \\ &= \text{Trace}\left[\left(\sum_{b=1}^{k}\mathbb{X}_b^T\mathbb{X}_b\right)^{-1}\right]\end{aligned}$$

Next, we take the expectation of the above expression w.r.t $\mathbb{X}_b$:

$$\mathbb{E}_{\mathbb{X}_b}\left[\text{Trace}\left[\left(\sum_{b=1}^{k}\mathbb{X}_b^T\mathbb{X}_b\right)^{-1}\right]\right] = \text{Trace}\left[\mathbb{E}_{\mathbb{X}_b}\left[\left(\sum_{b=1}^{k}\mathbb{X}_b^T\mathbb{X}_b\right)^{-1}\right]\right]$$

To simplify this, note that we have shown previously that $\mathbb{E}\left[\frac{\mathbb{X}_b^T\mathbb{X}_b}{n_b}\right] = \mathrm{I} + \boldsymbol{\mu_b}\boldsymbol{\mu_b}^T$. Therefore,

$$\begin{aligned}\mathbb{E}\left[\sum_{b=1}^{k}\mathbb{X}_b^T\mathbb{X}_b\right] &= \mathbb{E}\left[\sum_{b=1}^{k}n_b\left(\frac{\mathbb{X}_b^T\mathbb{X}_b}{n_b}\right)\right] \\ &= \sum_{b=1}^{k}n_b\mathbb{E}\left(\frac{\mathbb{X}_b^T\mathbb{X}_b}{n_b}\right) \\ &= \sum_{b=1}^{k}n_b(\mathrm{I} + \boldsymbol{\mu_b}\boldsymbol{\mu_b}^T) \\ &= n\left(\mathrm{I} + \sum_{b=1}^{k}\frac{n_b}{n}\boldsymbol{\mu_b}\boldsymbol{\mu_b}^T\right)\end{aligned}$$

We take the inverse using the Continuous Mapping Theorem (CMT):

$$\mathbb{E}\left[\sum_{b=1}^{k}\mathbb{X}_b^T\mathbb{X}_b\right]^{-1} = \frac{1}{n}\left(\mathrm{I} + \sum_{b=1}^{k}\frac{n_b}{n}\boldsymbol{\mu_b}\boldsymbol{\mu_b}^T\right)^{-1}$$

Therefore,

$$\begin{aligned}\mathbb{E}\left[\text{Var}\left[\hat{Y}_M(\mathbf{x}_\star)\right]\right] &= \text{Trace}\left[\mathbb{E}_{\mathbb{X}_b}\left[\left(\sum_{b=1}^{k}\mathbb{X}_b^T\mathbb{X}_b\right)^{-1}\right]\right] \\ &= \frac{1}{n}\text{Trace}\left(\mathrm{I} + \sum_{b=1}^{k}\frac{n_b}{n}\boldsymbol{\mu_b}\boldsymbol{\mu_b}^T\right)^{-1}\end{aligned}$$

■

B.2. Lemma 9

We simplify the expected variances of the *Ensemble* and the *Merged* from Lemma 8 in the case in which there are two clusters within the training set.

Lemma 9 *Assume $k = 2$. We denote $\lambda_b = \frac{n_b}{n}$ for $b = 1, 2$; by construction, $\lambda_1 + \lambda_2 = 1$. Then, we have the following:*

(i)

$$\lim_{n\to\infty} \frac{\mathbb{E}\left[Var\left[\hat{Y}_E(\mathbf{x}_\star)\right]\right]}{s_1} = 1$$

(ii)

$$\lim_{n\to\infty} \frac{\mathbb{E}\left[Var\left[\hat{Y}_M(\mathbf{x}_\star)\right]\right]}{s_2} = 1$$

where

$$s_1 = \frac{1}{n}\left[(S-1) + \frac{(S-1)[1+\lambda_2\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}+\lambda_1\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}]+1}{(S-1)(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})+1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}+\lambda_2\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}\right]$$

$$s_2 = \frac{1}{n}\left[S - \frac{\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}} - \frac{\lambda_2\boldsymbol{\mu_2}^T\boldsymbol{\mu_2} - \frac{2\lambda_1\lambda_2}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}\boldsymbol{\mu_1}^T\boldsymbol{\mu_1} + \frac{\lambda_2\lambda_1^2}{(1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})^2}(\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})^2}{1+\lambda_2\boldsymbol{\mu_2}^T\boldsymbol{\mu_2} - \left(\frac{\lambda_1\lambda_2}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)\boldsymbol{\mu_2}^T\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T\boldsymbol{\mu_2}}\right]$$

Proof

(i) The expected variance of the *Ensemble* when k = 2 is:

$$\begin{aligned}
\mathbb{E}\left[\text{Var}\left[\hat{Y}_E(\mathbf{x}_\star)\right]\right] &= \left[n_1\left[(S-1)+\frac{1}{1+\|\boldsymbol{\mu_1}\|^2}\right]^{-1} + n_2\left[(S-1)+\frac{1}{1+\|\boldsymbol{\mu_2}\|^2}\right]^{-1}\right]^{-1} \\
&= \left[n\lambda_1\left[(S-1)+\frac{1}{1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right]^{-1} + n\lambda_2\left[(S-1)+\frac{1}{1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}\right]^{-1}\right]^{-1} \\
&= \frac{1}{n}\left[\frac{\lambda_1}{(S-1)+\frac{1}{1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}} + \frac{\lambda_2}{(S-1)+\frac{1}{1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}}\right]^{-1} \\
&= \frac{1}{n}\left[\frac{\lambda_1(S-1)+\frac{\lambda_1}{1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}+\lambda_2(S-1)+\frac{\lambda_1}{1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}}{\left[(S-1)+\frac{1}{1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right]\times\left[(S-1)+\frac{1}{1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}\right]}\right]^{-1} \\
&= \frac{1}{n}\left[\frac{\left[(S-1)+\frac{1}{1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right]\times\left[(S-1)+\frac{1}{1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}\right]}{\lambda_1(S-1)+\frac{\lambda_1}{1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}+\lambda_2(S-1)+\frac{\lambda_1}{1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}}\right]
\end{aligned}$$

The numerator of the above fraction is:

$$\begin{aligned}
&\left[(S-1)+\frac{1}{1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right]\times\left[(S-1)+\frac{1}{1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}\right]\\
&=(S-1)^2+\frac{S-1}{1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}+\frac{S-1}{1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}+\frac{1}{(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})}\\
&=\frac{(S-1)^2(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})+(S-1)[2+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}]+1}{(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})}\\
&=\frac{[S-1]\Big[(S-1)(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})+[1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}+\lambda_2\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}]+[1+\lambda_2\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}+\lambda_1\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}]\Big]+1}{(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})}
\end{aligned}$$

The denominator of the above fraction is:

$$\begin{aligned}
&\lambda_1(S-1)+\frac{\lambda_1}{1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}+\lambda_2(S-1)+\frac{\lambda_1}{1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\\
&=\frac{(S-1)(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})+\lambda_1(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})+\lambda_2(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})}{(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})}\\
&=\frac{(S-1)(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})+1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}+\lambda_2\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}{(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})}
\end{aligned}$$

Putting this all together:

$$\begin{aligned}
&\frac{1}{n}\left[\frac{\left[(S-1)+\frac{1}{1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right]\times\left[(S-1)+\frac{1}{1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}\right]}{\lambda_1(S-1)+\frac{\lambda_1}{1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}+\lambda_2(S-1)+\frac{\lambda_1}{1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}}\right]\\
&=\frac{1}{n}\left[\frac{[S-1]\Big[(S-1)(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})+[1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}+\lambda_2\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}]+[1+\lambda_2\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}+\lambda_1\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}]\Big]+1}{(S-1)(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})+1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}+\lambda_2\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}\right]\\
&=\frac{1}{n}\left[(S-1)+\frac{(S-1)[1+\lambda_2\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}+\lambda_1\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}]+1}{(S-1)(1+\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})(1+\boldsymbol{\mu_2}^T\boldsymbol{\mu_2})+1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}+\lambda_2\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}}\right]
\end{aligned}$$

(ii) To simplify the expected variance of the *Merged*, we begin by considering the expression $(\mathrm{I}+\lambda_1\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T+\lambda_2\boldsymbol{\mu_2}\boldsymbol{\mu_2}^T)^{-1}$. To simplify, we use the Sherman-Morrison Inverse formula, which states that for matrix $A_{S\times S}$ and vectors $u, v\in\mathbb{R}^S$,

$$(A+uv^T)^{-1}=A^{-1}-\frac{A^{-1}uv^TA^{-1}}{1+v^TA^{-1}u}$$

We apply the Sherman-Morrison formula twice in succession; for the first round, $A=\mathrm{I}+\lambda_1\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T$, $u=\lambda_2\boldsymbol{\mu_2}$, and $v=\boldsymbol{\mu_2}$:

$$(\mathrm{I}+\lambda_1\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T+\lambda_2\boldsymbol{\mu_2}\boldsymbol{\mu_2}^T)^{-1}=(\mathrm{I}+\lambda_1\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)^{-1}-\frac{(\mathrm{I}+\lambda_1\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)^{-1}\lambda_2\boldsymbol{\mu_2}\boldsymbol{\mu_2}^T(\mathrm{I}+\lambda_1\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)^{-1}}{1+\lambda_2\boldsymbol{\mu_2}^T(\mathrm{I}+\lambda_1\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)^{-1}\boldsymbol{\mu_2}}$$

We apply the formula again to the term $(\mathrm{I} + \lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)^{-1}$, now, $A = \mathrm{I}, u = \lambda \boldsymbol{\mu_1}$, and $v = \boldsymbol{\mu_1}$:

$$(\mathrm{I} + \lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)^{-1} = \mathrm{I} - \frac{\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1 + \lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}$$

We plug this into the previous expression:

$$\begin{aligned}
&(\mathrm{I} + \lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)^{-1} - \frac{(\mathrm{I} + \lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)^{-1}\lambda_2 \boldsymbol{\mu_2}\boldsymbol{\mu_2}^T(\mathrm{I} + \lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)^{-1}}{1 + \lambda_2 \boldsymbol{\mu_2}^T(\mathrm{I} + \lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)^{-1}\boldsymbol{\mu_2}} \\
&= \mathrm{I} - \frac{\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1 + \lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}} - \frac{\left(\mathrm{I} - \frac{\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1+\lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)\lambda_2 \boldsymbol{\mu_2}\boldsymbol{\mu_2}^T\left(\mathrm{I} - \frac{\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1+\lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)}{1 + \lambda_2 \boldsymbol{\mu_2}^T\left(\mathrm{I} - \frac{\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1+\lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)\boldsymbol{\mu_2}}
\end{aligned}$$

We now take the trace of the above expression, and use the linearity of the trace operator to first consider the trace of each of the component three terms. We furthermore use the fact that for vector ν, Trace$[\nu\nu^T] = \nu^T\nu$.
The first term:

$$\text{Trace}\,[\mathrm{I}] = S$$

The second term:

$$\begin{aligned}
\text{Trace}\left[\frac{\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1 + \lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right] &= \frac{\lambda_1}{1 + \lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\text{Trace}\left[\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T\right] \\
&= \frac{\lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}{1 + \lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}
\end{aligned}$$

For the third term, first denote the constant $1 + \lambda_2 \boldsymbol{\mu_2}^T\left(\mathrm{I} - \frac{\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1+\lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)\boldsymbol{\mu_2}$ as B:

$$\begin{aligned}
&\text{Trace}\left[\frac{\left(\mathrm{I} - \frac{\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1+\lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)\lambda_2 \boldsymbol{\mu_2}\boldsymbol{\mu_2}^T\left(\mathrm{I} - \frac{\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1+\lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)}{1 + \lambda_2 \boldsymbol{\mu_2}^T\left(\mathrm{I} - \frac{\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1+\lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)\boldsymbol{\mu_2}}\right] \\
&= \frac{1}{B}\text{Trace}\left[\left(\mathrm{I} - \frac{\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1 + \lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)\lambda_2 \boldsymbol{\mu_2}\boldsymbol{\mu_2}^T\left(\mathrm{I} - \frac{\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1 + \lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)\right] \\
&= \frac{1}{B}\text{Trace}\left[\lambda_2 \boldsymbol{\mu_2}\boldsymbol{\mu_2}^T\left(\mathrm{I} - \frac{\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1 + \lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)^2\right] \text{, by cyclic property of the Trace operator} \\
&= \frac{1}{B}\text{Trace}\left[\lambda_2 \boldsymbol{\mu_2}\boldsymbol{\mu_2}^T\left(\mathrm{I} - \frac{2\lambda_1 \boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1 + \lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}} + \frac{\lambda_1^2(\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)^2}{(1 + \lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1})^2}\right)\right] \\
&= \frac{1}{B}\left[\lambda_2 \boldsymbol{\mu_2}^T\boldsymbol{\mu_2} - \frac{2\lambda_1\lambda_2}{1 + \lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\text{Trace}\left[\boldsymbol{\mu_2}\boldsymbol{\mu_2}^T\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T\right] + \frac{\lambda_2\lambda_1^2}{(1 + \lambda_1 \boldsymbol{\mu_1}^T\boldsymbol{\mu_1})^2}\text{Trace}\left[(\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)^2\right]\right]
\end{aligned}$$

To simplify this above expression, we note the following identities:

$$\begin{aligned}
\text{Trace}\left[(\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)^2\right] &= \left(\text{Trace}\left[\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T\right]\right)^2 \\
&= (\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})^2
\end{aligned}$$

and similarly,

$$\begin{aligned}\text{Trace}\left[(\boldsymbol{\mu_2}\boldsymbol{\mu_2}^T)(\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)\right] &= \text{Trace}\left[(\boldsymbol{\mu_2}\boldsymbol{\mu_2}^T)\right]\text{Trace}\left[(\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T)\right] \\ &= \boldsymbol{\mu_2}^T\boldsymbol{\mu_2}\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}\end{aligned}$$

The third term then becomes:

$$\frac{1}{A}\left[\lambda_2\boldsymbol{\mu_2}^T\boldsymbol{\mu_2} - \frac{2\lambda_1\lambda_2}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}\boldsymbol{\mu_1}^T\boldsymbol{\mu_1} + \frac{\lambda_2\lambda_1^2}{(1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})^2}(\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})^2\right]$$

Finally, putting it all together,

$$\begin{aligned}&\frac{1}{n}\text{Trace}\left[(\mathrm{I}+\lambda_1\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T+\lambda_2\boldsymbol{\mu_2}\boldsymbol{\mu_2}^T)^{-1}\right] \\ &= \frac{1}{n}\text{Trace}\left[\mathrm{I} - \frac{\lambda_1\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}} - \frac{\left(\mathrm{I}-\frac{\lambda_1\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)\lambda_2\boldsymbol{\mu_2}\boldsymbol{\mu_2}^T\left(\mathrm{I}-\frac{\lambda_1\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)}{1+\lambda_2\boldsymbol{\mu_2}^T\left(\mathrm{I}-\frac{\lambda_1\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)\boldsymbol{\mu_2}}\right] \\ &= \frac{1}{n}\left[S - \frac{\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}} - \frac{\lambda_2\boldsymbol{\mu_2}^T\boldsymbol{\mu_2} - \frac{2\lambda_1\lambda_2}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}\boldsymbol{\mu_1}^T\boldsymbol{\mu_1} + \frac{\lambda_2\lambda_1^2}{(1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})^2}(\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})^2}{1+\lambda_2\boldsymbol{\mu_2}^T\left(\mathrm{I}-\frac{\lambda_1\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)\boldsymbol{\mu_2}}\right] \\ &= \frac{1}{n}\left[S - \frac{\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}} - \frac{\lambda_2\boldsymbol{\mu_2}^T\boldsymbol{\mu_2} - \frac{2\lambda_1\lambda_2}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\boldsymbol{\mu_2}^T\boldsymbol{\mu_2}\boldsymbol{\mu_1}^T\boldsymbol{\mu_1} + \frac{\lambda_2\lambda_1^2}{(1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})^2}(\boldsymbol{\mu_1}^T\boldsymbol{\mu_1})^2}{1+\lambda_2\boldsymbol{\mu_2}^T\boldsymbol{\mu_2} - \left(\frac{\lambda_1\lambda_2}{1+\lambda_1\boldsymbol{\mu_1}^T\boldsymbol{\mu_1}}\right)\boldsymbol{\mu_2}^T\boldsymbol{\mu_1}\boldsymbol{\mu_1}^T\boldsymbol{\mu_2}}\right]\end{aligned}$$

■

B.3. Proof of Theorem 3

Proof We consider the case in which $k = 2$ and $\|\boldsymbol{\mu_1}\| = \|\boldsymbol{\mu_2}\| = 1$ (that is, $\boldsymbol{\mu_1}^T\boldsymbol{\mu_1} = \boldsymbol{\mu_2}^T\boldsymbol{\mu_2} = 1$), and simplify the quantities in Lemma 9.

(i) $\mathbb{E}\left[\text{Var}\left[\hat{Y}_E(\mathbf{x}_\star)\right]\right]$ becomes:

$$\begin{aligned}S - 1 + \frac{(S-1)[1+\lambda_1+\lambda_2]+1}{4(S-1)+1+\lambda_1+\lambda_2} &= S - 1 + \frac{2(S-1)+1}{4(S-1)+2} \\ &= S - \frac{1}{2}\end{aligned}$$

(ii) $\mathbb{E}\left[\text{Var}\left[\hat{Y}_M(\mathbf{x}_\star)\right]\right]$ becomes:

$$S - \frac{\lambda_1}{1+\lambda_1} - \frac{\lambda_2 - \frac{2\lambda_1\lambda_2}{1+\lambda_1} + \frac{\lambda_1^2\lambda_2}{(1+\lambda_1)^2}}{1-\lambda_2-\frac{\lambda_1\lambda_2}{1+\lambda_1}} = S - \frac{1+2\lambda_1\lambda_2 - \frac{3\lambda_1^2\lambda_2+2\lambda_1\lambda_2}{1+\lambda} + \frac{\lambda_1^2\lambda_2+\lambda_1^3\lambda_2}{(1+\lambda_1)^2}}{2+\frac{\lambda_1^2\lambda_2-\lambda_1\lambda_2}{1+\lambda_1}+\lambda_1\lambda_2}$$

Now, we show that second term above is larger than $\frac{1}{2}$, and therefore that the expected variance of the *Merged* is lower than that of the *Ensemble*. Let $a = 1+2\lambda_1\lambda_2 - \frac{3\lambda_1^2\lambda_2+2\lambda_1\lambda_2}{1+\lambda} + \frac{\lambda_1^2\lambda_2+\lambda_1^3\lambda_2}{(1+\lambda_1)^2}$ (the numerator), and $b = 2 + \frac{\lambda_1^2\lambda_2-\lambda_1\lambda_2}{1+\lambda_1} + \lambda_1\lambda_2$ (the denominator). Showing that $\frac{a}{b} > \frac{1}{2}$ is equivalent to showing that $b - 2a < 0$.

$$\begin{aligned} b - 2a &= \frac{7\lambda_1^2\lambda_2 + 3\lambda_1\lambda_2}{1+\lambda_1} - 3\lambda_1\lambda_2 - 2\left[\frac{\lambda_1^2\lambda_2 + \lambda_1^3\lambda_2}{(1+\lambda_1)^2}\right] \\ &= \frac{(7\lambda_1^2\lambda_2 + 3\lambda_1\lambda_2)(1+\lambda_1) - 3\lambda_1\lambda_2(1+\lambda_1)^2 - 2[\lambda_1^2\lambda_2 + \lambda_1^3\lambda_2]}{(1+\lambda_1)^2} \\ &= \frac{-4\lambda_1^2\lambda_2 + 2\lambda_1^3\lambda_2 - 6\lambda_1\lambda_2}{(1+\lambda_1)^2} \\ &< 0 \end{aligned}$$

Therefore,

$$S - \frac{1 + 2\lambda_1\lambda_2 - \frac{3\lambda_1^2\lambda_2+2\lambda_1\lambda_2}{1+\lambda} + \frac{\lambda_1^2\lambda_2+\lambda_1^3\lambda_2}{(1+\lambda_1)^2}}{2 + \frac{\lambda_1^2\lambda_2-\lambda_1\lambda_2}{1+\lambda_1} + \lambda_1\lambda_2} < S - \frac{1}{2}, \text{ and thus}$$

$$\lim_{n\to\infty} \frac{\mathbb{E}\left[\text{Var}\left[\hat{Y}_M(\mathbf{x}_\star)\right]\right]}{\mathbb{E}\left[\text{Var}\left[\hat{Y}_E(\mathbf{x}_\star)\right]\right]} < 1$$

■

B.4. Proof of Theorem 4

In this section, we derive limiting expressions for Equations 1a and 1b when S grows proportionally with n. Throughout, we denote by $\boldsymbol{Z} \overset{\text{i.i.d.}}{\sim} \text{N}_S(\mathbf{0}, \text{I}_n)$ the $n \times S$ matrix corresponding to the standard normal component of $\mathbb{X}$, by $\boldsymbol{Z_t} \overset{\text{i.i.d.}}{\sim} \text{N}_S(\mathbf{0}, \text{I}_{n/2})$ the $\frac{n}{2} \times S$ cluster-specific standard normal matrix for $t = 1, 2$, by $\mathbf{z}_i$ the p-dimensional observation-specific standard normal vector for $i = 1, \ldots, n$, and and by $\bar{\mathbf{z}}$ the sample mean of the $\mathbf{z}_i$'s. Throughout, we use that if $\boldsymbol{Z} \overset{\text{i.i.d.}}{\sim} \text{N}_S(\mathbf{0}, \text{I}_n)$, the inverse sample covariance matrix $(\boldsymbol{Z}^T\boldsymbol{Z})^{-1}$ follows an Inverse Wishart distribution with parameters n and I_p. The first moment of the $W^{-1}(n, \text{I})$ distribution is $E[W^{-1}] = \frac{1}{n-S-1}\text{I}_S$. Therefore,

$$\begin{aligned} \lim_{n,p\to\infty} \text{Trace}\left(E\left[\left(\boldsymbol{Z}^T\boldsymbol{Z}\right)^{-1}\right]\right) &= \lim_{n,p\to\infty} \frac{1}{n-S-1}\text{Trace}\left[\text{I}_S\right] \\ &= \frac{\gamma}{1-\gamma} \end{aligned}$$

We use the above results in the following proof.

Proof

(i) We begin by considering the *Merged* learner. Given our assumptions, the merged data $\mathbb{X} = [\mathbb{X}_1, \mathbb{X}_2]^T$, comprises of n i.i.d observations following a mixture model:

$$\begin{aligned} \mathbf{x}_i &\sim AF_1 + (1-A)F_2 \\ \text{where } A &\sim \text{Bernoulli}(1/2) \\ F_1 &\sim N(\boldsymbol{\mu}, I) \\ F_2 &\sim N(-\boldsymbol{\mu}, I) \end{aligned}$$

From this, we can derive the mean and variance of each observation $\mathbf{x}_i = \mathbf{z}_i + \boldsymbol{\mu}, i = 1, ..., n$, yielding

$$\begin{aligned} E[\mathbf{x}_i] &= 0 \\ \text{Var}[\mathbf{x}_i] &= \mathrm{I} + \boldsymbol{\mu}\boldsymbol{\mu}^T \end{aligned}$$

Thus, the sample covariance matrix $\frac{\mathbb{X}^T\mathbb{X}}{n}$, converges to $\mathrm{I} + \boldsymbol{\mu}\boldsymbol{\mu}^T$ as $n \to \infty$. The variance of the *Merged* is equal to the identity matrix plus a rank-1 perturbation, which only affects the largest eigenvalue. We denote the perturbation as ζ_1.

We use this information to then take the limit of the *Merged* risk:

$$\begin{aligned} 1 + \mathbf{x}_\star^T \left(\mathbb{X}^T\mathbb{X}\right)^{-1} \mathbf{x}_\star &\to 1 + \mathbb{E}_{\mathbf{x}_\star}\left[\mathbf{x}_\star^T \left(\mathbb{X}^T\mathbb{X}\right)^{-1} \mathbf{x}_\star\right] \\ &= 1 + \frac{1}{n}\text{Trace}\left[\left(\frac{\mathbb{X}^T\mathbb{X}}{n}\right)^{-1}\right] \\ &= 1 + \frac{1}{n}\sum_{j=1}^{p} \hat{\lambda}_j + \frac{\zeta_1}{n} \\ &= 1 + \frac{1}{n}\text{Trace}\left[\left(\frac{\boldsymbol{Z}^T\boldsymbol{Z}}{n}\right)^{-1}\right] + \frac{\zeta_1}{n} \\ &\to 1 + \frac{1}{n}\text{Trace}\left[\left(\frac{\boldsymbol{Z}^T\boldsymbol{Z}}{n}\right)^{-1}\right] \\ &= 1 + \text{Trace}\left[\left(\boldsymbol{Z}^T\boldsymbol{Z}\right)^{-1}\right] \\ &\to 1 + \text{Trace}\left[\mathbb{E}\left[\left(\boldsymbol{Z}^T\boldsymbol{Z}\right)^{-1}\right]\right] \\ &= 1 + \frac{\gamma}{1-\gamma} \end{aligned}$$

where $\hat{\lambda}_j$ are the eigenvalues of $\left(\frac{\boldsymbol{Z}^T\boldsymbol{Z}}{n}\right)^{-1}$.

(ii) To analyze the *Ensemble*, we use the same notation as above and note that we can write the cluster-specific unscaled inverse covariance matrix as follows:

$$\left(\mathbb{X}_b^T\mathbb{X}_b\right)^{-1} = (\boldsymbol{Z}_\mathbf{b}^T\boldsymbol{Z}_\mathbf{b} + \boldsymbol{\mu}\boldsymbol{\mu}^T + 2\boldsymbol{\mu}_\mathbf{b}\bar{\mathbf{z}}^T)^{-1}$$

for $b = 1, 2$. Using the Sherman-Morrison-Woodbury formula, we get that

$$\left(\mathbb{X}_b^T \mathbb{X}_b\right)^{-1} = (\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b} + \boldsymbol{\mu}\boldsymbol{\mu}^T)^{-1} - \frac{2(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b} + \boldsymbol{\mu}\boldsymbol{\mu}^T)^{-1}\boldsymbol{\mu}_\mathbf{b}\bar{\mathbf{z}}^T(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b} + \boldsymbol{\mu}\boldsymbol{\mu}^T)^{-1}}{1 + 2\bar{\mathbf{z}}^T(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b} + \boldsymbol{\mu}\boldsymbol{\mu}^T)^{-1}\boldsymbol{\mu}_\mathbf{b}}$$

and recall from above that

$$\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b} + \boldsymbol{\mu}\boldsymbol{\mu}^T\right)^{-1} = \left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-1} - \frac{\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-1}\boldsymbol{\mu}\boldsymbol{\mu}^T\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-1}}{1 + \boldsymbol{\mu}^T\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-1}\boldsymbol{\mu}}$$

Then, the quantity

$$\begin{aligned}
\mathbb{E}_{\mathbf{x}_\star}\left[\mathbf{x}_\star^T\left(\mathbb{X}_b^T \mathbb{X}_b\right)^{-1}\mathbf{x}_\star\right] &= \text{Trace}\left[\left(\mathbb{X}_b^T \mathbb{X}_b\right)^{-1}\right] \\
&= \text{Trace}\left[(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b} + \boldsymbol{\mu}\boldsymbol{\mu}^T)^{-1} - \frac{2(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b} + \boldsymbol{\mu}\boldsymbol{\mu}^T)^{-1}\boldsymbol{\mu}_\mathbf{b}\bar{\mathbf{z}}^T(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b} + \boldsymbol{\mu}\boldsymbol{\mu}^T)^{-1}}{1 + 2\bar{\mathbf{z}}^T(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b} + \boldsymbol{\mu}\boldsymbol{\mu}^T)^{-1}\boldsymbol{\mu}_\mathbf{b}}\right] \\
&= \text{Trace}\left[\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-1}\right] - \text{Trace}\left[\frac{\boldsymbol{\mu}^T\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-2}\boldsymbol{\mu}}{1 + \boldsymbol{\mu}^T\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-1}\boldsymbol{\mu}}\right] \\
&\quad - 2 \times \text{Trace}\left[\frac{\bar{\mathbf{z}}^T(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b} + \boldsymbol{\mu}\boldsymbol{\mu}^T)^{-2}\boldsymbol{\mu}_\mathbf{b}}{1 + 2\bar{\mathbf{z}}^T(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b} + \boldsymbol{\mu}\boldsymbol{\mu}^T)^{-1}\boldsymbol{\mu}_\mathbf{b}}\right]
\end{aligned}$$

Let $\hat{\delta}_j$ and $\hat{v}_j$ be the jth eigenvalue and eigenvector, respectively, of $\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-1}$, for $j = 1, \ldots, p$. We will need the following result in order to simplify the expression above.

$$\begin{aligned}
\boldsymbol{\mu}^T\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-2}\boldsymbol{\mu} &= \sum_{j=1}^{S} \hat{\delta}_j^2\left(\boldsymbol{\mu}^T \hat{v_j}\right)^2 \\
&= \sum_{j=1}^{S} \hat{\delta}_j^2\left(\frac{1}{\sqrt{S}}\sum_{l=1}^{p}\boldsymbol{\mu}^{(l)}\right)^2 \\
&= \frac{1}{S}\left(\sum_{l=1}^{S}\boldsymbol{\mu}^{(l)}\right)^2 \sum_{j=1}^{S}\hat{\delta}_j^2 \\
&= \frac{1}{S}\text{Trace}\left[\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-2}\right] \\
&\to \frac{1}{S}\text{Trace}\left[E\left[\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-2}\right]\right] \\
&= 0
\end{aligned}$$

Therefore, the terms Trace $\left[\frac{\boldsymbol{\mu}^T\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-2}\boldsymbol{\mu}}{1+\boldsymbol{\mu}^T\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-1}\boldsymbol{\mu}}\right]$ and Trace $\left[\frac{\bar{\mathbf{z}}^T(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}+\boldsymbol{\mu}\boldsymbol{\mu}^T)^{-2}\boldsymbol{\mu}_\mathbf{b}}{1+2\bar{\mathbf{z}}^T(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}+\boldsymbol{\mu}\boldsymbol{\mu}^T)^{-1}\boldsymbol{\mu}_\mathbf{b}}\right]$ both converge to 0 as $n, S \to \infty$; when expanded, both expressions contain the term $\boldsymbol{\mu}^T\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-2}\boldsymbol{\mu}$ in the numerator.

Given the above results,

$$\begin{aligned}
\mathbb{E}_{\mathbf{x}_\star}\left[\mathbf{x}_\star^T\left(\mathbb{X}_b^T \mathbb{X}_b\right)^{-1}\mathbf{x}_\star\right] &\to \text{Trace}\left[\left(\boldsymbol{Z}_\mathbf{b}^T \boldsymbol{Z}_\mathbf{b}\right)^{-1}\right] \\
&= \frac{\gamma}{\frac{1}{2} - \gamma}
\end{aligned}$$

and therefore the limiting expected *Ensemble* risk is

$$\lim_{n,p\to\infty} \mathbb{E}_{\mathbf{x}_\star}\left[1+\sum_{i=1}^{2}\mathbf{x}_\star^T\left(\mathbb{X}_b^T\mathbb{X}_b\right)^{-1}\mathbf{x}_\star\right] \to 1+2\left(\frac{\gamma}{\frac{1}{2}-\gamma}\right)$$

Taking the ratio of (i) and (ii) completes the proof. ■